Useful possibility collapse identifies adaptive emergent structure in learning systems

[Authors]
[Affiliations]

Abstract

Emergence is used to describe abrupt abilities, collective coordination, representational phase changes, synergy and macro-causal efficacy, yet these signatures often disagree. We propose an intervention-aware framework that identifies *adaptive emergence* in learning systems as *useful possibility collapse*: the endogenous formation and stabilization of a valuable macro-structure in an open conditional future distribution. A known trajectory-space substrate fixes what is measured; the output is a six-component certificate plus a calibrated continuous record whose dimensions demonstrably measure the constructs they name against a ground-truth generator. The same frozen thresholds identify learned systems and reject scripted and initialization controls in three built domains and—under an externally timestamped preregistration—in the public Overcooked-AI benchmark, where all five registered predictions passed. Multivariate combinations of prior measures with equal freedom fail to transfer to fresh systems, and a future-distribution score prospectively discovers value-critical decisions in uncurated human chess. Emergence becomes testable rather than curve-read.

"Emergence" may be the most load-bearing and least agreed-upon word in the science of complex and learning systems [1, 2, 3, 4, 5, 6]. Four communities measure it four ways: abrupt performance curves in large language models [8, 9, 10, 11, 7, 22, 23], sudden representation jumps, informational synergy [15, 16, 17, 14, 18, 21], and macro-level causal efficacy under intervention [13, 19, 20].

These are not four estimators of one agreed-upon quantity but distinct operational definitions that yield conflicting verdicts: Wei et al. [7] call LLM abilities emergent; Schaeffer et al. [12] show the same curves can be artifacts of the metric and sampling grid, and no performance curve carries information about where an ability came from. Synergy and causal-emergence measures quantify macro-structure without encoding whether it was selected by the system, useful in context, or acquired rather than imposed. Here we organize these signatures on systems with an explicit trajectory law and value structure: each audited quantity is a deterministic functional of the same trajectory-law family (Proposition 4), yet none of them—alone (Proposition 1), in exact published form on fully enumerated chains, or in fitted multivariate combination—reproduces the predeclared labels on a controlled battery, because no projection retains the value sign, the conditional selectivity, and the provenance of the collapse together.

What, then, do we propose to measure? The claim is deliberately scoped: we do not offer an observer-free definition of all emergence in nature, but an identification framework for *adaptive emergence acquired during learning*. Its layers are explicit. A system holds an *open possibility space*—many future basins with non-trivial probability; an *endogenous trigger*—an action or learning event chosen by the system, often locally costly—collapses that distribution *selectively* (conditioned on context) and *specifically* (triggered futures differ from non-triggered ones). This much is *structural collapse*, and it can occur in harmful or prewired systems. The event qualifies as adaptive emergence only when the collapse is also *useful* (a positive value do-contrast, assessed through model-based interventions [43]), *endogenous* and *acquired* rather than prewired. In one sentence: *adaptive emergence is the stabilized, useful collapse of an* *acquired behavioural structure in an open possibility space.* Six components—potential, selectivity, specificity, usefulness, endogeneity, acquisition—make this falsifiable; thresholds were frozen before the confirmatory measurements to which they applied. The layering preserves what a single conjunction would flatten: harmful collapse and scripted coordination are structurally emergent but not adaptive, and both cases are measured systems in our batteries rather than rhetorical categories.

The mathematical starting point is honest about its lineage. The trajectory-space object we use is a standard specific-information quantity, the same posterior–prior divergence that defines Bayesian surprise [44, 45]; related future-facing quantities—empowerment as an action–future channel capacity [54, 55], causal path-entropy forces [56], and semantic information as viability-relevant syntactic information [46]—have long used information and intervention to separate valuable from incidental structure. What these frameworks do not supply, and what this paper adds, is an identification protocol for *when* a selective, valuable, acquired contraction of the future distribution should count as emergence: the substrate is known mathematics; the contribution is the layered criterion, its falsification battery, and its cross-domain transfer under frozen thresholds.

The paper then does four things. *First*, it shows why the prior signatures conflict: on a controlled battery each is a lossy projection of one trajectory-law family, and named imitations fail exactly the components they are built to fail—neither single prior signals nor their fitted multivariate combinations transfer to fresh systems. *Second*, it fixes what is reported: the primary output is a multi-dimensional record (certificate, causal magnitude, signed value, abruptness, acquisition, persistence, uncertainty), with scalar summaries declared secondary; the record's dimensions are calibrated against a ground-truth generator and satisfy eight machine-verified axioms. *Third*, it validates: the full six-component conjunction confirms under frozen thresholds in three built system families—an external swarm, Contextual Level-Based Foraging, and a latent-context sequence model—and, under an externally timestamped preregistration, in the public Overcooked-AI benchmark. *Fourth*, it shows what the account buys beyond postdiction: a derived phase boundary forecasts where emergence appears in a parametrized family before training; an event-level score performs prospective discovery in uncurated chess play; and the process proxy recovers grokking [26, 27] and induction-head formation [36, 37] and selected transitions in public MultiBERTs [38] and Pythia [39] series while rejecting gradual controls. Registered failures are retained and delimit scope, including the martingale lesson that converged policies price their own triggers into the baseline, so useful collapse in that regime is visible only through do-contrasts.

1 Results

A trajectory-space framework for emergence

The substrate: a known information quantity. Let τ be a system trajectory with prior law $P(\tau)$—the open possibility space—and let M record which basin, ability or outcome materialized. The trajectory-space information associated with observing structure m is

$$C(m) \;=\; \mathrm{KL}\big(P(\tau \mid M{=}m) \,\big\|\, P(\tau)\big) \qquad \text{[bits]}. \tag{1}$$

Equation (1) is not new mathematics and is not, by itself, a definition of emergence: it is the specific-information form of Bayesian surprise [44, 45], and it depends only on how rare the realized macro-outcome is ($\mathbb{E}_m C(m) = I(\tau; M)$; for deterministic readouts $C(m) = -\log_2 P(A_m)$). Its role here is to fix the *space* on which emergence claims must live—distributions over futures, not points in representation space or scores on a metric. Three statistics that the word "collapse" could conflate are connected by an exact bridge:

Proposition 0 (Collapse bridge)**.** *At a declared step with action distribution π and a fixed rollout law thereafter (so that choosing a and* $\mathrm{do}(a)$ *induce the same future law), the action-attributable prospective contraction $H(P_t(B)) - \mathbb{E}_{a\sim\pi} H(P(B \mid \mathrm{do}(a)))$, the π-averaged interventional divergence $\mathbb{E}_{a\sim\pi}\mathrm{KL}(P(B \mid \mathrm{do}(a)) \,\|\, P_t(B))$, and the expected retrospective update $\mathbb{E}_b\mathrm{KL}(\pi(A \mid B{=}b) \,\|\, \pi(A))$ all equal $I(A;B)$. A deterministic policy has $I(A;B) = 0$.*

The bridge separates system-driven collapse from ordinary information revelation: every resolving stochastic process concentrates its posterior, but only the contraction attributable to the system's own action choice is common to all three readings, and it vanishes for deterministic controllers—which is why greedy decisions carry no potential, and why all measurements below use the prospective, interventional forms (identities verified to machine precision on 10,000 random systems and on trained trigger steps; Methods).

The substrate also shows why it cannot carry the verdict alone: on the fully enumerated battery chains, raw path-space KL is large in controls too (converged team 32.2 bits, shaped process 31.9 bits), and only the declared value-bearing basin projection separates the accepted system (32% of its trajectory-level contrast retained, against 3% and 0.01% for controls). The discriminating content lives in the observer contract—a basin partition $B = g(\tau)$ that resolves the candidate macro-structure, a value function, a rollout policy, interventions and a time resolution—and in the component criteria built on it. This is the architecture of the framework, not a limitation discovered after the fact: known substrate, declared contract, layered criteria.

Layered identification criteria. A collapse event is scored under a declared observer contract in two layers:

Definition 1 (Structural collapse)**.** An episode-level event is a structural collapse iff it has (1) **Potential**, an open future before the event; (2) **Selectivity**, context-dependent rather than unconditional collapse; and (3) **Specificity**, triggered futures that differ from non-triggered futures.

Definition 2 (Adaptive emergence)**.** A structural collapse qualifies as adaptive emergence iff it also has (4) **Usefulness**, a positive value do-contrast; (5) **Endogeneity**, a trigger chosen by the system rather than imposed; and (6) **Acquisition**, a gain over the system's own initialization.

The two layers are not decoration: harmful collapse (a decoy trap) and scripted coordination are structural collapses that fail the adaptive layer, and both are measured systems below. The layering also predicts an asymmetry that we test directly: structural components are properties of the future distribution and should survive a change of observer contract that still resolves the macro-structure, whereas usefulness is value-relative *by declaration* and may not.

Where six components are unavailable, we use scoped instruments instead: chess is an event-level test without acquisition, non-contextual deep MARL is a mechanism-level do-contrast test, and checkpoint series use a separate four-component process proxy in which burstiness measures acquisition velocity, not emergence magnitude (Table 2). The six components are identification criteria motivated by the substrate and validated by falsification below; they are not claimed to be uniquely derivable from Eq. (1).

The exclusions are empirical, not stipulated: deterministic greedy controllers have no open future; gradual frequency learning fails the burst component of the process proxy; scripted, forced and prewired systems fail provenance; memorization and harmful traps collapse possibilities without useful do-contrast. The verdict is a *conjunction* for a measured reason: a large factor would otherwise buy back a deficient one. Thresholds and analysis windows were frozen in code and per-domain author-maintained protocols before each confirmatory measurement; throughout, "registered" means *internally frozen* in these hash-anchored records—no protocol was third-party registered and no external timestamp is claimed, with one exception: the Overcooked-AI confirmation protocol carries a public timestamp that predates every confirmatory measurement (Methods).

Figure 1: Four regimes of the future-basin distribution, and one measured identification case. **a**, The observer's future-basin distribution $P_t(B \mid s_t)$ separates regimes that performance or representation jumps conflate: forced convergence fails Potential; structureless openness fails Selectivity and Usefulness; harmful collapse is structural but not adaptive (fails Usefulness); useful possibility collapse passes all components, and the do-blocked counterfactual (dotted) shows the future would otherwise have stayed open. **b**, Where each component is read off, on one stored Contextual LBF case (fresh seed 1101): per-context trigger rates against the initialization twin give selectivity (0.96 vs 0.01) and acquisition (+0.95); potential, do-contrast specificity and the value do-contrast clear their frozen thresholds (dashed); and the layered checklist accepts the learned policy while rejecting the twin, a competent scripted controller (fails exactly the adaptive layer) and a fixed rule.

Figure 1 gives the conceptual picture: four regimes of $P_t(B \mid s_t)$ that performance or representation jumps conflate—forced convergence (the future was never open), structureless openness (open but never collapses), harmful collapse (a structural collapse that fails the adaptive layer), and useful possibility collapse. Only the last is adaptive emergence under Definition 2, and each of the other three is a measured system in our battery, not a hypothetical.

Prior signatures diverge on a controlled battery

Proposition 1 (Single-observable insufficiency on the audited battery)**.** *For each tested scalar observable $O \in \{$potential $H(P_0)$; collapse burst; specificity JS; usefulness gap; representation jump; causal-emergence EI$\}$, no one-sided threshold reproduces all predeclared labels on the measured battery. The best hindsight-selected threshold accuracies are* 0.800–0.900.

The result follows by exhaustive threshold sweep, with measured witnesses in Methods (Table 3). Representation distance illustrates all three failure modes at once: the harmful decoy has the *same* jump (0.102) as the true conditional emergence (usefulness +3.43 vs −8.39; a norm carries no sign), the genuine case jumps *less* than reward-shaped (0.403) and converged-team (0.328) controls, and a noisy learner with the same selective structure jumps 13.7× farther. Collapse upper-bounds the jump (Proposition 2): a jump certifies that the distribution moved, never where it landed nor who moved it.

On the designed battery the component conjunction scores 1.000 against predeclared labels (Fig. 2a); after one registered selectivity failure and a disclosed refinement, the frozen rule confirms on a fresh internal seed (10/10) and on 25/25 fresh external swarm verdicts with measured acquisition (Methods). Dropping selectivity, usefulness or endogeneity re-admits its named counterexample; every single prior observable, even at its hindsight-optimal threshold, caps at 0.800–0.900 (Fig. 2a,b).

Exact rival formalisms within declared candidate families. Because charitable proxies could misrepresent the rival definitions, we also computed Hoel's EI/CE and Rosas' Ψ exactly, with zero Monte-Carlo error, on fully enumerated policy-closed Markov chains of all ten systems (up to 75,004 states)—exact evaluations *within predeclared restricted candidate families*, not exhaustive searches over all coarse-grainings or features. Exact CE (over a five-member coarse-graining family containing the mechanistically privileged partitions) is negative for every system; its best threshold is the trivial all-negative classifier (accuracy 0.8, missing both positives). Exact Ψ (maximized over four supervenient features and two micro decompositions) scores 0.3 under its published sign rule, its top scorer is the wrong-selector system (+0.59), and its hindsight optimum requires inverting the published sign convention while still missing the true conditional emergence (Fig. 2c). The blind spots—no access to value sign, natural selectivity, or provenance—are structural on this testbed.

A graded measurement from the rarity identity. The substrate also predicts a gradient no audited signature supplies: the less the training signal biases search toward a macro-structure, the rarer that structure is under the provenance's early search distribution ($C = -\log_2 P$). Measured for one macro-pattern under three provenances, the seed-mean ordering is as predicted—scripted $0 <$ process-shaped $0.39 <$ outcome-only 0.65 bits, with discovery twice as late at matched final competence—while outcome rarity and structural magnitude are provenance-indistinguishable by construction, so the three quantities a single "strength" number would conflate are reported separately. The registered suddenness ordering failed and is retained: both acquisitions are step-like, so this is a provenance-adjusted discovery surprise, not a universal magnitude (Extended Data Fig. 13).

Harmful emergence and matched provenance. Two constructions make the layering load-bearing (nine registered predictions, all passed). A policy trained on an exploiter reward *learns* a harmful structure on 5/5 seeds: it selectively lures its partner onto the decoy in exactly the harmful context, passes every structural component, and its value do-contrast is +7.5 under the declared beneficiary's value while -2.0 under the team value—structurally emergent under both values, adaptive only for its beneficiary. And four systems with matched natural behaviour (script, behavioural clone, shaped and outcome-only learners; pattern probability ≥ 0.99 for all) separate what a provenance flag alone could not: the do-contrast distinguishes the clone from its source (specificity 0.38–0.45 vs 1.0 bits—forcing the clone off its training distribution exposes states it never memorized); acquisition separates static from trained provenance; and provenance rarity orders the trained routes (clone $0.33 <$ shaped $0.39 <$ outcome-only 0.65 bits).

A continuous record under the certificate. The binary certificate prevents dimensional compensation; a continuous record answers *how strongly, how abruptly and in which direction*. Each system receives a declared vector—normalized potential, selectivity, causal magnitude, signed value, acquisition, abruptness—whose aggregation was fixed before calibration and never fitted to labels. That these dimensions are measured constructs rather than plausible design is established against ground truth: a six-knob generator whose parameters independently control each construct by construction, measured through the same finite-sample estimator pipeline as the real systems, yields a diagonally dominant sensitivity matrix with no violation (Extended Data Fig. 15); the value knob spans 1.52 of V while moving M by less than 0.001, and the acquisition knob spans 0.99 of Q at unchanged structure. Nullity, value separability, provenance separability and dose–response monotonicity (Spearman 1.0, 5/5 seeds) passed as registered;

the off-diagonal suppression rule failed on two couplings and is retained (one exemption-list specification error; persistence responds to structure at 0.29 of its diagonal because retention is retention *of* structure). Eight axioms of the record are machine-verified, and observer freedom is itself a computable quantity: across the stored admissible contracts, per-seed structural scores carry identification intervals of median width 0.14 with a stable ranking (mean pairwise Spearman 0.76), and no control's interval touches the adaptive layer. The record's honest boundary is predictive: three registered early-prediction rules failed and are retained—structure-only scores cannot separate scripted from learned (the layering's own claim, re-derived), and near phase boundaries early causal magnitude does not beat early performance at predicting acceptance, though each dimension does predict its *own* endpoint (early magnitude forecasts final structure at Spearman 0.56 where performance is uninformative at −0.09). The predictive content is axis-specific, not a universal improvement over performance.

Multivariate prior-signal baselines with equal freedom. Comparing one scalar against a six-threshold conjunction would be asymmetric, so we gave the prior signals equal or greater freedom and tested generalization rather than fit. Hindsight AND-rules over pairs or triples of prior signals, logistic regression and a depth-2 tree on all signals (including exact EI and Ψ) reach 0.9–1.0 in-sample, drop to 0.7–0.8 under leave-one-out CV, and—frozen and applied to a fresh-seed battery on which the frozen six-component protocol had already scored 10/10—transfer at 0.7–0.9, with every baseline misclassifying the true conditional emergence; our own two-component AND-rules transfer no better (0.7–0.8). What the projections lack is not fitting capacity but the selectivity, value and provenance information they discard (Fig. 2d).

Possibility collapse within single episodes

The definition is about futures, so it is measured *while the future is still open*: within single episodes we estimate $P_t(B \mid s_t)$ by rollouts and apply do-operators at the trigger step (Extended Data Fig. 9). The same action yields positive return gaps in the rescue context and negative gaps in the bridge (decoy) context—the directional content no unsigned jump carries. The basin partition is not hand-picked: label-free clustering recovers the designed basins with purity 1.00 in all displayed regimes (one registered partial failure on pure-team loss episodes is retained). A neural replication reproduces the sign structure, and neural embedding jumps track collapse bursts (correlation 0.87–0.90, peak alignment 1.0)—the representation jump appearing exactly where Proposition 2 says it must, as a downstream symptom.

Proposition 2 (Collapse bounds representation jump)**.** *Let* $R_t = \mathbb{E}_{B\sim P_t}[\phi(B)]$ *for a feature map* ϕ*,* $J_t = \|R_t - R_{t-1}\|_2$*, and* $K_t = \mathrm{KL}(P_t \,\|\, P_{t-1})$ *in bits. Then*

$$J_t \;\le\; \mathrm{diam}(\phi)\,\mathrm{TV}(P_t, P_{t-1}) \;\le\; \mathrm{diam}(\phi)\sqrt{\tfrac{\ln 2}{2}\,K_t}. \tag{2}$$

Burst-like collapse *can* produce representation jumps—which is why jump-based observables fire when collapse happens—but the converse fails: a large J_t certifies only that the distribution moved, not that the movement was a useful collapse. The bound was verified numerically at every step of every trajectory in four training regimes and both grokking conditions.

Proposition 3 (Counterfactual necessity credits selectivity)**.** *Let a system trigger with probability* p_c *in context* c *(contexts drawn with probability* w_c*), and let* $V_c(a)$ *be the expected return under* $a \in \{\textit{trigger}, \textit{non-trigger}\}$*. With* $U = \mathbb{E}[\textit{return} \mid \textit{behavior}] - \mathbb{E}[\textit{return} \mid \mathrm{do}(\textit{non-trigger})]$*,*

$$U \;=\; \sum_c w_c\, p_c\big(V_c(\textit{trigger}) - V_c(\textit{non-trigger})\big). \tag{3}$$

A selective system ($p_c \approx 1$ where the trigger helps, ≈ 0 where it harms) receives the full positive terms and none of the negative ones: U measures exactly the value of the selection structure. A blind system receives the sign of the *marginal* effect—which is why blind triggering can show positive U in environments where triggering helps on average, and why usefulness alone cannot replace selectivity. Equation (3), applied with the closed-form V_c of a parametrized environment, is also what powers the within-family prospective parameter forecast of Section 1.

Process probes detect acquisition in public models

A definition of emergence in learning systems must also account for acquisition over training. Here the event is a checkpoint transition rather than a do-blockable action, so we use the separate process proxy from Table 2: potential before the window, a collapse burst in the future-basin distribution, usefulness gain on a fixed evaluation battery and controls that share either collapse or performance symptoms. This proxy measures acquisition *velocity*; it is not a stand-alone emergence verdict.

The proxy first recovers two cases with independently studied mechanisms (Extended Data Figs. 3, 4, 10). Grokking transitions in three architecture families pass the frozen potential, burstiness and usefulness thresholds, while the memorization phase in the same run is rejected: held-out entropy collapses ($6.6 \to 2.8$ bits) with no accuracy gain. The two-layer attention-only model that can implement induction passes all components (copy accuracy 0.987); the one-layer architecture that cannot fails usefulness (0.112).

On public series the same protocol transfers as a resolution-limited instrument. MultiBERTs subject–verb agreement collapses from a 14.7-bit masked-verb space in a burst coinciding with the accuracy jump $0.50 \to 0.98$ on all five published seeds, with paired controls rejected; Pythia decoders repeat the accept/reject double dissociation at four scales while both frequency-tail families are rejected at every scale. The held-out 2.8B run marks the boundary: usefulness passes but burstiness fails ($3.2 < 5$) because collapse is spread across several early checkpoints, and every checkpoint thinning flips the verdict back (retained S7 failure)—checkpoint spacing makes abruptness non-identifiable there, a measured resolution limit, not the absence of a transition. An ordinary supervised learner is accepted by the proxy in 6/6 runs, confirming it as an acquisition-shape instrument, never a stand-alone emergence verdict. Hash audits of all public checkpoints, including two upstream defects detected and quarantined at 2.8B, back these numbers (Extended Data Figs. 10, 8).

Confirmation and parameter extrapolation in multi-agent systems

Mechanistic transfer in deep MARL. Multi-agent systems are a canonical setting for unprogrammed coordination [31, 32, 33, 34, 35]. In PettingZoo simple_spread, trained MAPPO policies hold 1.4–1.6 bits of early potential (untrained and noise policies are even more open and correctly fail to qualify), and forcing one agent toward its eventual assignment rather than blocking it shifts the bijection-basin probability positively in every policy seed. Because the training-population unit is the seed, we report seed-level inference (Fig. 4a): all six simple_spread seeds are positive (sign test $p = 0.016$; cluster-bootstrap mean $[0.032, 0.156]$), and all eight Level-Based Foraging seeds are positive ($p = 0.004$; $[0.159, 0.390]$), with the three-seed registered runs and the post-hoc extensions distinguished throughout. The registered win-shift prediction failed in both tasks, reproducing the martingale lesson. A detector comparison including a hand-coded cooperator shows why performance or structure alone is not a verdict: the scripted cooperator matches or exceeds learned policies on win rate, specificity, Ψ and EI, but is rejected by provenance.

Full six-component confirmation across domains. The first full-criterion external transfer, in the continuous swarm benchmark, classified 25/25 fresh verdicts. We then froze a Contextual LBF protocol after two disclosed pilots. The environment's dynamics, observations, actions and sparse rewards are unchanged; the wrapper only balances two unlabeled geometric contexts. Nine of ten learned policies pass all six components; the retained miss is seed 1104 (conditional selectivity $0.4875 < 0.5$). All ten learned policies have the predicted context ordering and positive usefulness/acquisition, all 40 initialization/scripted controls reject, and the competent team-nearest controller passes all behavioural components while failing exactly endogeneity and acquisition on every seed. Five additional post-confirmation seeds support the same direction (4/5 learned passes, 20/20 controls rejected) but are not pooled with the frozen result (Fig. 3a,b).

This domain also carries the observer, rollout and model-error audits; we state their conclusions here and their designs, numbers and retained misses in Methods and Extended Data. The verdict does not need a hand-made possibility space: machine-partitioned and cross-fitted basins (four clustering methods on low-level trajectories only) reject 60/60 controls under every method and agree with the hand-basin protocol on 95.7% of verdicts, and 13,000 adversarial random partitions manufacture zero acceptances. A single *semantics-free* recipe—bag-of-opaque-tokens plus raw numeric summaries, standardized, k-means with k chosen by silhouette, byte-identical code and hyperparameters—then reproduces the verdicts across structurally unrelated domains: 8/9 systems on the gridworld battery with both predeclared positives accepted, 18/18 verdicts and 15/15 control rejections in the collective-control domain, alongside the stored 95.7% CLBF agreement. Anyone can compute the possibility space; the one battery disagreement (a control whose potential the semantics-free partition inflates by resolving contract-irrelevant micro-variation) is the residual observer-dependence living exactly where the refinement guarantee says it must. The verdict is not an artifact of one observer: under a second plausible contract and a seven-contract ensemble, all control verdicts are invariant negatives (60/60 and 420/420), the structural layer agrees on 14/15 learned seeds, and every learned flip is conservative and concentrated in the declaredly value-relative usefulness component (Extended Data Fig. 12)—structural collapse travels across reasonable contracts, adaptive qualification is relative to the declared value, exactly as the definition states. Openness is not decoding noise (near-greedy rollouts keep every learned seed above threshold; one diffuse-model prediction failed and is retained), and verdicts from learned world models replacing the simulator are reliable when training data is adequate—with the registered ensemble-disagreement decidability rule failing on shared bias (retained) and a disclosed coverage-augmented follow-up eliminating every silent wrong verdict at the cost of conservatism.

A sequence-domain full-criterion test. To test the protocol beyond embodied RL, we froze a latent-context sequence task for a small causal transformer trained only by next-token prediction: the context is never labelled, the valued continuation rule differs by context, decoding interventions force or forbid the first continuation token, and an initialization twin supplies acquisition. Across ten fresh seeds, learned transformers pass all six components (10/10), all 40 controls reject, and seed-bootstrap lower bounds are positive for usefulness (0.481) and acquisition (0.918) (Fig. 3c). One route prediction failed informatively: the deterministic oracle router is intervention-inert, so it fails specificity and usefulness in addition to provenance. A frozen generalization audit reproduces the boundary pattern of Contextual LBF; marker positions outside the training range collapse selectivity. The sequence domain is a controlled next-token setting, not a claim about natural language.

A public third-party environment, preregistered externally. Every environment above was built or configured by us. The strongest remaining objection—that the criterion works only in author-designed worlds—is answered in Overcooked-AI, a public, unmodified coordination benchmark [34]. The full protocol (layout pair, training recipe, trigger, basins, both observer contracts, thresholds copied unchanged from the frozen criterion, and five predictions OC-1–OC-5) was frozen with a *public external timestamp* before any confirmatory seed was launched; the run script refuses to start without asserting that the timestamp exists. Twelve self-play PPO seeds were then trained on a mixed pair of layouts (5×10^6 steps each), with four controls per seed. All five registered predictions passed (Extended Data Fig. 14): learned policies are accepted on all six components on 8/12 seeds (registered line 8/12); all 48 control verdicts are rejections, with the layered failure routes—the scripted role pair and its behavioural clone are competent but fail endogeneity and acquisition, twins and untrained pairs fail selectivity and usefulness; the trigger direction matches the pilot on 12/12 seeds; the declared second contract never rescues a twin (12/12); and the primary seed-level inference is unambiguous, a positive usefulness do-contrast on 12/12 seeds (exact sign test $p = 2 \times 10^{-4}$). The four rejected learned seeds fail selectivity—they pot in both layouts at similar rates—which is the correct verdict for a policy that did not develop context-selective structure, not a miss of the protocol. The continuous record, assembled read-only with a declared value scale, agrees without seeing the verdicts: every accepted seed has a positive adaptive score, all 48 controls sit at zero, and the four rejected seeds rank strictly below every accepted seed on structural score (0.653 max vs 0.695 min, no overlap).

Collective control. A supplementary family reconstructs the structure of crowd-scale collective control (the Twitch-Plays-Pokémon phenomenon: thousands of humans steering one avatar invented a context-dependent vote-aggregation convention) in a system where every component is measurable: forty noisy voters drive one avatar, the system's action is the aggregation mode itself, and a meta-controller is trained on value alone. All 50 control verdicts reject with exactly the declared routes—blanket majority-voting fails potential (forced convergence), blanket anarchy fails usefulness through falls (the historical counterfactual, reproduced by do-block on 10/10 seeds), and the hand-scripted switcher and its clone fail exactly endogeneity and acquisition. Two registered predictions failed and are retained (7/10 learned acceptances against a registered 8/10, all three rejections being context-blind blanket democracy; graded rather than binary selectivity), and two design pilots are disclosed (Supplementary Information). The historical event itself is observational-scale: no do-operator or twin exists, exactly the scope boundary Table 2 declares.

The rejections themselves obey the framework. In two independent domains, roughly a third of seeds learn a context-*blind* convention and fail exactly selectivity—so we froze a bifurcation battery asking whether the blind convention is a lawful competing attractor. Manipulating the cost of the coordinated mode over five levels (ten fresh seeds each), the selective-seed fraction moves from 0.1 to 0.9–1.0 across the point where the *measured* value gap between the two conventions (computed from scripted policies, no learner involved) changes sign, and the gap's sign predicts the majority basin at 4/5 grid points—the single miss being the transition region itself (gap +0.12). Applying the usefulness algebra of Proposition 3 to convention selection thus predicts *which* training outcomes the certificate will reject, before training. One registered prediction failed and is retained: an early-training snapshot does not predict the final convention basin (chance level; the basin is decided late), so the bifurcation is forecastable at the population level but not, at this resolution, for individual seeds. A follow-up battery locates the decision (five further misses retained across it and its disclosed margin re-measurement; Supplementary Information): at the population level the convention space never collapses—fifty seeds settle into a stable 26/22/2 split, a bifurcation rather than a universal attractor—and the deciding variable is not hazard learning at all. Both conventions end with strong, indistinguishable preferences at the hazard state; on 24/24 seeds with no overlap, what separates them is the *open-ground* state, where selective seeds discover that the cheap uncoordinated mode suffices and blind seeds never do. The certificate's selectivity rejections thus track a late, stochastic discovery event in the low-stakes context—measured, not hypothesized.

The framework forecasts. Definitions postdict; mechanisms predict. Applying Proposition 3 with the closed-form payoffs of the parametrized gridworld yields derived boundaries *before training*: behavioural onset of the trigger at goal reward $G = 5$, usefulness sign change at $G = 9$, second onset at $G = 11$ (boundaries registered in advance). Independent learners trained at $G \in \{3, 5, 7, 9, 11, 13, 16\}$ then reproduced the derived phase structure: no structure for $G \leq 5$; a selective-but-harmful phase at $G = 7$—the wrong-selector pattern arising *naturally* from payoffs, rejected via usefulness as the framework requires; and accepted emergence above the $G = 9$ analytical usefulness boundary. The primary registered run matched all four non-tie grid points; three independent-seed replications matched 11/12 non-tie points (91.7%), with one false acceptance at $G = 7$. The outer phases were stable (reject at $G \leq 5$, accept at

$G \geq 13$ in 3/3 seeds); registered tie points $G \in \{5, 9, 11\}$ were not scored. A two-dimensional extension derives a *non-rectangular* acceptance surface over goal reward and context mix before training—acceptance requires both a sufficient payoff and a sufficient context frequency—and 80 independent learners (16 cells $\times$ 5 seeds) match the derived verdict on 14/15 non-fragile cells by seed majority, reproducing the predicted interaction (within G=10 the verdict flips along the context axis; within low context weight it flips along the payoff axis). The registered all-cells prediction failed by one cell (G=13, context weight 0.25: 2/5 seeds acquire the structure; low context frequency makes discovery itself stochastic) and is retained. The experiments support a forecasted transition region rather than an exact noise-free boundary (Fig. 4b).

Prospective discovery in uncurated chess play

Chess supplies candidate events and an evaluator not produced by the authors [29, 30]: human annotations nominate costly candidate key moves, and an engine supplies the counterfactual future distribution. On 240 externally annotated sacrifices and 120 quiet controls under an internally frozen protocol, the key moves show the event-level signature (Fig. 5a). They are *locally costly* (median -3.0 pawns against the best reply; 80.0% $[74.6, 85.0]$ strictly costly), *counterfactually decisive* ($P(\text{win} \mid \text{do}(\text{key})) - P(\text{win} \mid \text{do}(\text{best-alternative}))$ mean $+0.539$ $[0.508, 0.570]$, positive in 240/240 positions), and *not mere openness*: quiet positions are more open ($+0.764$ bits) yet no single move usefully collapses them—the opportunity is a property of the state–action pair. All core conclusions hold in 12/12 robustness cells over temperature, depth, basin thresholds and a classical engine (Extended Data Fig. 11). One frozen prediction failed informatively: the behaving policy's future does not improve after the key move because the engine prices the move into its own baseline—$P_t(\text{win})$ is a martingale—and the identical failure appeared independently in the deep-MARL win-shift prediction above, forcing the do-contrast principle stated in the Introduction.

Recovery is not discovery, so a separately frozen protocol sampled 400 uncurated positions per month from public dumps (population filters only), scored them with the shallow observer alone (potential and the future-basin do-gap between the depth-4 first and second candidate moves), and labelled them afterwards with a deep-search referee that never feeds back into the scores. All four frozen predictions passed in both months (Fig. 5b): the do-gap ranks value-critical decisions at AUROC 0.730 and 0.725, beats tactical (0.635) and material (0.589) baselines, and the frozen flag lifts precision 2.5$\times$ and 3.0$\times$ over base rate with median flagged potential 1.9 bits. Because movers repeat within a month, inference is also reported at the mover-cluster level: the cluster bootstrap gives AUROC intervals $[0.615, 0.829]$ and $[0.613, 0.834]$, wider than position-level intervals but excluding chance by a clear margin. The engine-native eval-gap baseline is not distribution-stable ($0.762 \to 0.652$ across months) while the do-gap is unchanged. Four referee-family checks bound the remaining circularity, two passing and two retained as informative failures. A classical (pre-NNUE) referee leaves the signal intact (AUROC 0.743 and 0.669), and mover-cluster bootstrap intervals exclude chance. An engine-free realized-outcome referee is directionally consistent (humans play the shallow-best move far more often at flagged positions, 0.53 vs 0.34 and 0.43 vs 0.27, with positive realized-score gains) but its registered interaction is an underpowered null (permutation $p = 0.54$; one decision 20–60 plies before the end is washed out by subsequent play at these rating bands). A different-lineage engine referee (Toga II, Fruit family) fails the frozen transfer rule (0.663 and 0.534; retained): its compressed evaluation scale yields 3–5 positives per month and degrades every predictor including the engine-native baseline, so the check cannot separate predictor non-transfer from weak-referee label noise. A lineage-independent referee of comparable strength remains open (`CHESS_DISCOVERY_PREREGISTRATION.md`).

Prior emergence signatures as low-dimensional projections

Proposition 4 (Coverage and insufficiency of audited signatures)**.** *Let $\mathcal{P} = \{P(\tau \mid c, \mathrm{do}(w))\}$ be the intervened family of trajectory laws over contexts c and interventions w (including none), together with the observer's declared value, representation, macro-feature and micro-decomposition maps. Then:* (i) Derivability. *Each audited emergence quantity— performance jump (Wei et al.), representation jump, Hoel's exact EI and CE, Rosas' exact Ψ, and PID synergy—is a deterministic functional $Q = F_Q(\mathcal{P})$, obtained by coarse-graining τ and/or restricting the intervention set, then applying a fixed information-theoretic or metric map.* (ii) Audited verdict insufficiency. *None of these scalar functionals, under either its published sign rule or a hindsight-optimal one-sided threshold, reproduces all predeclared labels on the battery. This is an empirical statement about the declared testbed and candidate families, not a universal ranking theorem.*

Derivability alone is weak—any statistic computed from a sufficiently rich family is a functional of it. The content is the conjunction of (i) with (ii) and the transfers above, and the framework predicts *when* each signature agrees or fails: performance jumps are bounded symptoms of collapse (Proposition 2), with grader-dependence exactly Schaeffer et al.'s critique [12]; EI intervenes uniformly, so it cannot see which states the system itself selects and is sign-blind to value; Ψ reads behavioural marginals, so forced or wrongly-selective coordination scores highly (its exact top scorer is the wrong-selector); and when a converged policy prices its trigger into the baseline, observational value change is a martingale while the do-contrast remains nonzero—observed independently in chess and deep MARL. Table 1 summarizes each projection and its measured blind spot.

Each projection also carries a measured *agreement condition* under which it coincides with the full verdict: performance jumps agree when value gain, selectivity and provenance co-occur (the learned positives); representation jumps agree when the moved distribution is also a useful collapse (Prop. 2 witnesses the converse failures); Ψ agrees when coordination is both learned and correctly context-gated; EI-style efficacy agrees when the macro-partition is the verdict-carrying one. The battery's counterexamples are exactly the systems that break each condition while satisfying the others. The scope of the claim is precise: *covered* means derivable-and-audited on systems with an explicit trajectory law and value structure, within the declared candidate families. We do not claim that Ψ or EI are wrong in their home use cases (quantifying macro predictive structure, or intervention-level compression); the claim is that *as emergence verdicts* they are projections that discard the verdict-carrying information—the value sign, the conditional selectivity, and the provenance of the collapse—and that the layered episode criterion adds structure that no audited projection, singly or in fitted combination, retains.

2 Discussion

The evidence is not flat, and the paper's weight rests on a stated hierarchy. *Decisive*: the controlled battery with exact rival forms and equal-freedom baselines; the externally timestamped Overcooked-AI confirmation; the pre-derived phase boundaries; the prospective chess discovery. *Supporting*: the three built full-criterion families, the generator calibration and axioms, the observer and provenance audits. *Boundary-setting*: the process proxy's grid-resolution failures, the persistence and generalization limits, the contract-relative value verdicts. *Failed and retained*: thirty-six registered predictions, none repaired in place. A reader who accepts only the decisive tier already has the paper's claim; the rest raises confidence and marks edges.

The substrate (1) is agnostic to the domain of the possibility space, and two instantiations cover the field's classic intuitions. *Temporal collapse*: a locally costly or silent act whose meaning arrives later, when it proves counterfactually necessary—chess sacrifices are the purest case (local cost -3 pawns, do-contrast $+0.54$), and the MultiBERTs NPI result (collapse at 20k, usefulness at 40k) shows the same shape at training scale. *Spatial collapse*: a joint basin no individual can reach

Table 1: Prior emergence signatures as projections of $P_t(B)$, each with a measured blind spot. All numbers from our runs; rival thresholds are hindsight-optimal (best case for the rival). Chess percentages: fraction of 240 externally annotated positions where the signature ranks the key move first.

Prior signature	**What it reads off $P_t(B)$**	**Measured blind spot (witness)**
Performance jump [7]	usefulness only, no provenance	accepts the reward-shaped system; misses latent conditional ability (hindsight acc. 0.800)
Representation jump	projection bounded by collapse (Pinsker) (Prop. 2)	direction-, provenance-, and strength-blind: harmful decoy same jump as accepted positive; noise 13.7× larger; acc. 0.900; chess 5.4%
PID synergy / Ψ [14] (*exact* Ψ computed)	joint structure of the collapsed state	exact $\Psi > 0$ scores 0.3; top scorer is the wrong-selector (+0.59); best threshold must invert the published sign and still misses the conditional positive
Causal emergence, EI [13] (*exact* EI computed)	macro intervention efficacy	exact CE < 0 on all 10 enumerated chains; best threshold = trivial classifier (0.8), misses both true positives; proxy's top scorer was the reward-*shaped* system
Metric-artifact critique [12]	grid-resolution sensitivity of jumps	*confirmed* from the mechanism side: our tail-facts results show abruptness is metric- and grid-dependent; burst is a measurement anchor, not a definition component

or hold alone, certified by single-agent do-blocks degrading the joint outcome—the operational content of "the whole exceeds the parts" without the synergy machinery the exact-form audit shows cannot carry a verdict. The duality is presentation, not new theory: both are contractions of one future distribution.

The spatial case carries an exact identity that formalizes two folk intuitions and one measured blind spot (machine-verified; Methods). The joint contraction relative to the independence null, $C_{\text{spatial}} = \sum_i H(B_i) - H(B_1, \ldots, B_N)$, *is* the total correlation of the agents' futures [47, 52], the quantity integration-style measures build on [53]. Populations feel more emergent because the open possibility space grows linearly with N under independence, so the collapse available to coordination scales with the group; and coordination is candidate emergence precisely because it contracts the joint future space until each individual is predictable from the others. But the identity also delimits: a deterministic script attains the *maximal* total correlation, and two mechanistically different generators with the same joint law are indistinguishable—structural collapse is provenance-blind as a matter of mathematics, which is why synergy-flavoured detectors accept scripted coordination on our battery and why the adaptive layer, not a better structure statistic, is what separates them.

The same vocabulary classifies the field's textbook exemplars, with five frozen expectations all met (Methods). In Boids flocking [48, 49], within-episode heading contraction and the measured three-bird total correlation both rise monotonically with the alignment coupling ($0.00 \to 1.64$ and $0.04 \to 2.70$ bits), and decoupling the alignment channel mid-episode shifts the final basin law by a full bit at high coupling while doing nothing at zero coupling—structural collapse, counterfactually load-bearing, but *not* adaptive emergence: the dynamics are prewired and nothing is acquired. Schelling segregation [50] shows the same class with a tipping signature: the do-freeze contrast is load-bearing only near the tipping threshold (0.33 bits at $\tau = 0.7$, zero below), because below it the system reaches its basin before the intervention can matter. And Conway's Life [51] marks the substrate boundary exactly: fifty replays of one soup give one future ($H(\text{final} \mid \text{init}) = 0$; the future was never open), and with no action channel the action-attributable collapse is zero by Proposition 0—gliders are deterministic pattern formation, outside the adaptive scope for a stated reason rather than by fiat.

These reductions assemble into an explicit type lattice with per-dimension *analytic* ground truth, guarding against the circularity of validating an instrument only on its own generator. Four universal coordinates—non-additivity N (co-information, analytic on logic gates: XOR exactly 1 bit, majority-3 exactly 0.434), interaction dependence D (structure loss under marginal-preserving surrogate decoupling), causal autonomy A (macro-minus-micro effective information, exact on enumerated chains), and robustness R (recovery ratio after irrelevant perturbation)—define *weak emergence* as their joint satisfaction, with *causal* ($A > 0$), *adaptive* (acquired; this paper's certificate) and *functional* (signed value) tiers above it, and philosophical strong emergence declared outside the empirical framework rather than treated as a higher score. Thresholds were calibrated on Boolean gates, exact Markov chains and constructed processes only, then frozen and applied blind to families the calibration never saw: supercritical Kuramoto oscillators classify as weak emergence and subcritical ones reject, Conway-glider dynamics classify as weak but not adaptive—exactly Chalmers' canonical assignment—and a stored learned convention classifies as adaptive (4/4). An eight-row adversarial matrix (common-driver synchrony, central controllers, redundant copying, transient coincidences, static patterns, thresholded pseudo-abilities, harmful congestion) fails each imposter on the predicted dimension, with harmful congestion correctly *accepted* as weak emergence with negative value (8/8). Three rounds of specification errors in the battery's own truth tables were caught by its exact computations before any blind system was scored, and are retained in the ledger.

The closest existing frameworks share parts of this architecture. Semantic information [46] also combines intervention with a viability-style value, but scores the information a system *has about* its environment; our criterion scores a *structure the system forms*, adding contextual selectivity, explicit future-basin collapse, acquisition against an initialization twin, and a cross-domain protocol with frozen thresholds. Empowerment [54, 55] and causal path-entropy forces [56] value *keeping* futures open; this framework identifies when a selective, valuable *contraction*, acquired during learning, should be recognized as adaptive emergence. Bayesian surprise [44] supplies the substrate quantity itself; none of its uses attach provenance, selectivity or value direction, which is where the layered criteria live.

Of the ∼259 internally frozen predictions, forty-six failed (nineteen before this revision, plus the realized-outcome interaction, the second-contract value flips, the strength-gradient suddenness ordering, the weak different-lineage referee, the diffuse rollout model, one two-dimensional forecast cell, the bias-blind world-model error proxy, the constant-series calibration rule, the structure-only class-separation rule, the early-magnitude prediction battery, the generator's off-diagonal rule, two convergent-validity endpoints, the two crowd-domain misses, the bifurcation snapshot rule, the meta-collapse commitment rules, and the universal-recipe, blind-demonstration and coordinate-battery specification misses reported here); none were discarded or re-thresholded post hoc. The failures carry the paper's sharpest lessons. Abruptness proved metric- and grid-dependent, confirming the metric-sensitivity critique [12] from the mechanism side and consistent with quantized-computation accounts of scaling [28]; two failures independently exposed the martingale principle for converged systems; and the acquisition component itself was added after a random network passed marginal selectivity by accident, then excluded the untrained control on every subsequent fresh seed.

The contribution is a scoped measurement framework, not an observer-free metaphysical definition. The full protocol is validated where every component is measurable (swarm, Contextual LBF, sequence model); the battery supplies targeted exclusions and exact rival-formalism evaluations within declared candidate families; chess supplies event-level recovery and prospective discovery but no acquisition; public checkpoints supply a resolution-limited process instrument, never a full verdict. We do not claim that all emergence in nature is possibility collapse, that prior definitions are special cases of ours, that frontier-scale language emergence is solved, or

Table 2: Scope of each empirical instrument. “Full” is reserved for settings in which all six components are measurable.

Evidence family	Instrument	Scope and claim supported
Controlled gridworld	five informative episode components	Action do-contrasts; forced constructions lack an informative initialization twin. Supports exclusion and threshold audit.
External swarm	full six-component criterion	Action do-contrasts and initialization twins. Supports fresh-seed full-criterion confirmation.
Contextual LBF	full six-component criterion	Balanced latent geometry, action do-contrasts, initialization twins and ten independent policy seeds; five additional post-confirmation seeds reported separately. Supports second-domain confirmation and robustness.
Latent-context sequence model	full six-component criterion	Next-token-trained transformer; decoding do-interventions and initialization twins; ten fresh seeds. Supports third-domain confirmation in the sequence modality.
Non-contextual deep MARL	potential + counterfactual usefulness	Action do-contrasts across policy seeds; not a full verdict. Supports cross-task mechanistic transfer.
Human chess	event-level strategic test	Forced engine alternatives; no acquisition measure. Supports external candidate/event transfer.
Grokking and public checkpoints	four-component process proxy	No action-level intervention; uses training provenance. Supports acquisition-process transfer.

that the proxy adjudicates “true” abruptness. Acquisition deliberately restricts the verdict to structure acquired during learning: prewired coordination and architecture-induced transitions can be structural collapses under Definition 1 but sit outside the adaptive verdict, and harmful collapse is representable rather than denied. The battery labels are designed diagnostic controls, not third-party ground truth; most frozen protocols are author-maintained, hash-anchored records without external timestamps, and one confirmation—in a public, unmodified third-party environment—carries an external timestamp that predates every measurement and passed all five of its registered predictions. Under declared, auditable observer contracts, useful possibility collapse can be measured, falsified and transferred, with structural verdicts stable across plausible contracts and value verdicts explicitly contract-relative. The collapse studied here is statistical—a contraction of the rollout future distribution driven endogenously by the system’s own actions or training dynamics—and carries no quantum connotation.

For machine intelligence, the framework turns “did an ability emerge?” from a curve-reading exercise into an intervention-aware measurement programme: a common trajectory-law substrate with measured blind spots for each prior projection, a conservative layered protocol where triggers, values, interventions and initialization twins are available, a separate velocity audit for public checkpoint series, a prospective-discovery test on uncurated data, and a negative-control practice in which frozen failures are retained and delimit scope. This is the sense in which possibility collapse unifies the audited uses of emergence: not by erasing their differences, but by placing them on one future-distribution object and showing which additional evidence turns a collapse into an adaptive-emergence verdict.

Methods

Scope of each empirical instrument

Table 2 maps every evidence family to the instrument it supports and the claim it can carry.

Proofs of the root identities

(0a) Expand the definition: $\sum_m P(m) \sum_\tau P(\tau \mid m) \log \frac{P(\tau|m)}{P(\tau)} = \sum_{m,\tau} P(m,\tau) \log \frac{P(m,\tau)}{P(\tau)P(m)} = I(\tau; M)$. *(0b)* If M is deterministic with cells A_m, then $P(\tau \mid A_m) = P(\tau)/P(A_m)$ on A_m and 0 elsewhere; substituting, the log-ratio is the constant $-\log_2 P(A_m)$ and the conditional mass sums to 1. *(0c)* Apply the data-processing inequality to the deterministic channel $g : \tau \mapsto B$; JS is a f-divergence-based mixture of KLs and inherits the inequality. Numerical verification: `verify_theory_bounds.py` (MI identity gap

$< 10^{-9}$; rarity law exact to $< 10^{-9}$; basin JS $0.90 \le$ trajectory JS 1.00 on do-contrast rollouts). The spatial-collapse identity (C_{spatial} = total correlation = KL to the product of marginals, ≥ 0; independence gives exactly linear growth of the joint entropy in N; coupling monotonicity; script-attained maximum $(N-1)\log_2 k$ and same-law generator indistinguishability) is machine-verified on 20,000 random joint laws and exact enumerated families (`verify_spatial_bridge.py`). The canonical-exemplars battery (`canonical_exemplars.py`) uses minimal standard implementations: 30 mixing headings with alignment weight and Gaussian noise (Boids; 3,000 episodes per coupling, 6-sector discretization for the empirical three-bird total correlation, do-decouple at mid-episode); a 20×20 torus with 35% vacancy and Moore-neighbourhood tolerance (Schelling; 800 episodes per threshold, do-freeze at one-sixth horizon); and Life on a 16×16 torus (400 random soups, 50 determinism replays of one soup). Expectations CE-1..CE-5 were frozen in the script docstring before running; all five passed.

Proof of Proposition 0

With a fixed rollout law after the declared step and no unobserved confounding (an MDP), $P(B \mid \mathrm{do}(a)) = P(B \mid A{=}a)$, so all three quantities are standard forms of the mutual information $I(A;B)$: the contraction form is $H(B) - H(B \mid A)$, the interventional form is $\mathbb{E}_a \mathrm{KL}(P(B \mid a)\|P(B))$, and the retrospective form is $\mathbb{E}_b \mathrm{KL}(\pi(A \mid b)\|\pi(A))$, equal by the symmetry of I. A deterministic π makes A constant, so $I(A;B) = 0$. The scientific content is the declaration that all manuscript measurements target the interventional form, with the identities verified numerically (10^4 random systems, max gap $< 10^{-15}$; trained gridworld trigger steps, gap $< 10^{-16}$; `verify_bridge_identity.py`).

Proof of Proposition 2

Write $\Delta(b) = P_t(b) - P_{t-1}(b)$, so $J_t = \|\sum_b \Delta(b)\phi(b)\|_2$. For any unit vector u, the scalar function $f_u(b) = u^\top \phi(b)$ has range at most $\mathrm{diam}(\phi)$. The total-variation duality gives

$$\left|\sum_b \Delta(b) f_u(b)\right| \le \mathrm{diam}(\phi)\,\mathrm{TV}(P_t, P_{t-1}).$$

Taking the supremum over unit u yields $J_t \le \mathrm{diam}(\phi)\mathrm{TV}(P_t, P_{t-1})$. The second step is Pinsker's inequality with the bit-to-nat conversion $\mathrm{KL}_{\mathrm{nats}} = \ln 2 \cdot K_t$. Verified at every step of every trajectory in four training regimes and both grokking conditions.

Observer refinement and model-error guarantees

Let B' refine B, with $B = h(B')$. Then $H(B') = H(B) + H(B'|B) \ge H(B)$, and data processing gives

$$\mathrm{JS}(P_a(B'), P_{a'}(B')) \ge \mathrm{JS}(P_a(B), P_{a'}(B)).$$

Thus a coarse-partition Potential or JS-Specificity pass remains a pass under refinement at the same absolute threshold. If value is coarse-measurable, $u'(b') = u(h(b'))$, its do-gap is exactly invariant. Entropy contraction itself is not refinement-monotone because the two conditional-entropy terms can change in either direction.

For rollout-model laws $Q_a, Q_{a'}$ approximating target laws $P_a, P_{a'}$, suppose

$$\mathrm{TV}(P_a, Q_a) \le \epsilon_a, \qquad \mathrm{TV}(P_{a'}, Q_{a'}) \le \epsilon_{a'},$$

and u has range R. Then

$$|\Delta_P u - \Delta_Q u| \le R(\epsilon_a + \epsilon_{a'}).$$

Accordingly, a measured usefulness margin larger than this bound is stable to the declared model error; without an estimate of the ϵ terms the do-gap remains observer-model relative. Finally, if each signed component margin exceeds its own error bound, the full conjunctive verdict is stable; a union bound gives confidence at least $1 - \sum_i \delta_i$. Ten thousand random-distribution checks for each partition/model bound and all 27 bounded-burst checks produced zero violations (`observer_bounds_verification.json`).

Witnesses for Proposition 1

The EI row uses the charitable episodic proxy; the exact-form computation (below) worsens the failure to the trivial classifier.

Table 3: Illustrative measured witness pairs: accepted and rejected systems that share, nearly share, or reverse a single-observable signal. Exhaustive threshold sweeps, rather than each pair alone, establish the best achievable battery accuracy.

Observable	S^+ (accepted)	S^- (rejected)	shared/reversed signal
potential	latent_conditional ($H_0 = 1.15$)	useful_habit ($H_0 = 1.11$)	open futures
collapse burst	grokking (burstiness 3974)	prewired (28376)	sudden collapse
specificity JS	noise_policy (0.74)	harmful_decoy (0.65)	trigger-contingent futures
usefulness gap	latent_conditional (+3.4)	useful_habit (+4.4)	positive gap
representation jump	uncertain-preference bridge	pure-team probe	large J_t
causal-emergence EI	latent_conditional (+0.71)	shaped_process (+0.93)	macro beats micro

Exact rival formalisms

For each of the ten battery systems we enumerated the policy-closed Markov chain exactly: state = (mode, context, agent positions, switch flag, clock), up to 75,004 states, with the learned softmax policy in closed form; the chain is time-inhomogeneous but exact. *Hoel EI:* $\mathrm{EI}(T) = I(X_{t+1}; X_t)$ with X_t uniform (max-entropy interventions) on the exact transition matrix; $\mathrm{CE} = \max_g \mathrm{EI}(\mathrm{macro}_g) - \mathrm{EI}(\mathrm{micro})$ over a five-member coarse-graining family including the mechanistically privileged partitions (mode-collapse, context-collapse, position-collapse, outcome classes, trivial). Exhaustive search over all coarse-grainings is super-exponential and is not performed in the original literature either; the conclusion is stated for this family. *Rosas* Ψ*:* $\Psi(V) = I(V_t; V_{t+1}) - \sum_j I(X_t^j; V_{t+1})$ on the pooled one-step joint of the behavioural occupancy measure, maximized over four supervenient features × two micro decompositions (the first-order practical criterion, as in the original applications; since verdicts are also evaluated at hindsight-optimal thresholds, constant redundancy offsets cannot rescue the ordering failures). Script: `exact_prior_formalisms.py`; output: `outputs/exact_prior_formalisms.json`.

Internal battery and refinement

Ten gridworld systems with designed ground truth; 36–96 rollouts per estimate. Plug-in entropy/JS estimation is consistent with Miller–Madow bias $O((|B| - 1)/n) < 0.09$ bits at the battery setting, below every margin the verdicts rely on. Estimator robustness: all qualitative conclusions unchanged across $n \in \{12..96\}$ and probe temperatures (12-cell grid); registered thresholds sit on wide accuracy plateaus (one honest edge documented: the noise policy sits at $H_0 = 0.508$, near the 0.5 potential cutoff). After a registered failure in which 2/5 external replication seeds accepted an untrained network on marginal tension, the criterion was refined *under freezing*: conditional selectivity $|p_{\mathrm{trig}}(A) - p_{\mathrm{trig}}(B)| \geq 0.5$; acquisition ≥ 0.3 separation gain over own initialization. The full six-component rule was confirmed out-of-sample on 25/25 external verdicts across five fresh seeds. A fresh internal seed classified 10/10 systems with the five informative components; acquisition was not scored internally because the forced-behaviour constructions have no informative initialization twin. The battery systems are designed diagnostic controls with predeclared labels, not an independent gold standard; their role is falsification pressure, and the independent tests are the frozen-threshold transfers, the public-checkpoint and chess studies, and the fresh-seed confirmations.

Fair multivariate baselines

Prior-signal baselines receive equal or greater freedom than the six-component conjunction: AND-rules over one to three prior signals (representation jump, metric jump, specificity, synergy, EI-style contrast; each signal with its own hindsight threshold and direction), an ℓ_2 logistic regression and a depth-2 decision tree on all five signals plus the exact EI and Ψ values, and two-component AND-rules over our own measured components. All models are fitted on the original battery scores, evaluated by leave-one-out cross-validation, then frozen and applied to a fresh-seed battery (seed 7011) whose five transportable prior signals were recomputed from scratch; the six-component reference on that seed is the stored confirmation, untouched by this analysis (`fair_baseline_comparison.py`, `fair_baseline_comparison.json`).

Grokking, induction heads, and scale

Grokking (modular addition, MLP + causal transformer + attention-only families): thresholds frozen in the bridge code (potential ≥ 1.0 bits; burstiness ≥ 5; usefulness gain ≥ 0.2; endogeneity by design flag); the memorization phase inside the same run is the sharpest internal control—held-out entropy 6.6 → 2.8 bits with zero accuracy gain, refused by the anchored window. This mirrors, in our vocabulary, the mechanistic progress measures that precede the grokking transition [27]: the collapse burst is a provenance-bearing progress measure, not a performance curve. Induction heads: attention-only transformers on cyclic-repeat sequences; the two-layer model passes all process-proxy components (copy accuracy 0.987), the one-layer model—which provably cannot implement the circuit [37]—fails usefulness (0.112); 12/12 registered predictions on fresh seeds. The in-context ability whose collapse we track is the same one shown to emerge only under bursty, heavy-tailed training distributions [24], consistent with our reading of it as an acquired—not prewired—collapse.

Public checkpoint series

MultiBERTs: 29 intermediate checkpoints × 5 seeds; 288 agreement minimal pairs; random-target controls share the agreement entropy/KL trace, while shuffled-vocabulary controls perturb the ability readout and test usefulness under the same public checkpoint series. Pythia-160m/410m/1B: 21 published checkpoints each, fetched from public Pythia checkpoint revisions; next-token distribution at the critical verb position; the same 288-pair construction (all verb pairs single-token in the GPT-NeoX vocabulary); tail families scored by mean per-token log-probability (eval-harness convention; tokenization amendment recorded before any checkpoint behaviour was observed). For both public series, the `train_acc` and `test_acc` columns store the same fixed evaluation-battery score for compatibility with the bridge analyser; there is no separate probe-training/test split, and no such independence is claimed. Conditions within a model seed share checkpoints and probe construction; the five model seeds provide seed replication, while paired controls diagnose verdict routes. All predictions and outcomes: `MULTIBERTS_PREREGISTRATION.md`, `PYTHIA_PREREGISTRATION.md` and `PYTHIA_SCALING_PREREGISTRATION.md`. The 1B and 1.4B runs use official sharded checkpoint revisions verified by a per-revision SHA-256 audit (all cached revisions pairwise distinct, `pythia_1.4b_checkpoint_hashes.json`); byte-range spot checks confirmed the download source serves content identical to `huggingface.co` for locally verified shards. The 2.8B run uses official sharded revisions after two upstream defects (a duplicated shard and a mismatched revision) were detected by the hash audit, quarantined and rebuilt from authentic sources; the earlier 2.8B mirror output remains quarantined and excluded.

Chess

240 positions selected by the Lichess `sacrifice` theme and rating band 1800–2400, plus 120 quiet controls; futures estimated by Stockfish rollout ensembles (registered temperature/depth); win/draw/loss basins by centipawn thresholds; do-operators implemented as forced first moves. Robustness grid: temperature × depth × basin thresholds × engine (incl. classical Stockfish 11). Protocol frozen in `CHESS_PREREGISTRATION.md`; the C5 margin failure and its martingale post-mortem are recorded there. Statistical intervals treat positions as independent units and are conditional on the curated annotation sample; no claim about population prevalence is made. The realized-outcome referee parses the same public PGN dumps for the game result and the human move actually played at each scored ply (never used by any scoring stage); the registered contrast is the realized-score gain from playing the shallow-best move at flagged versus unflagged positions, with a 10,000-shuffle permutation test on the interaction and bootstrap CIs per month; mover-rating balance between flagged and unflagged positions is reported as a confounder audit (`chess_realized_outcome.py`, `chess_realized_outcome.json`).

Deep MARL

`simple_spread`: MAPPO, 3 seeds, do-commit/do-block on target assignment; Level-Based Foraging (LBF, forced-cooperation `5x5-2p-2f`): MAPPO-style PPO, 3 seeds, sparse reward; interventions release on consumption (a pilot-stage semantic fix recorded with logs before the main run); probe temperature 6.0 chosen in pilots and stress-tested across $T \in \{2, 3, 4.5, 8\}$ with frozen success criteria. Detector comparison on 8 LBF systems includes forced-commit and a hand-coded cooperating imitation; the round-1 set-composition failure is archived and the round-2 prediction was frozen before the rerun. A post-hoc seed extension trained three further simple_spread and five further LBF policies with unchanged code and probes (`deep_marl_collapse_seed_extension.json`, `lbf_collapse_seed_extension.json`); combined seed-level statistics over 6 and 8 independent policies give exact one-sided sign tests $p = 0.016$ and $p = 0.004$ with positive cluster-bootstrap mean intervals $[0.032, 0.156]$ and $[0.159, 0.390]$ (`hierarchical_marl_analysis_combined.json`). The extension sizes were set by a seed-level power simulation from the three registered seeds (simulated one-sided power at $\alpha = 0.05$: ≥ 0.99 at $K = 5$ for simple_spread and ≥ 0.89 at $K = 5$, ≥ 0.99 at $K = 8$ for LBF), recorded in the same analysis file; the simulation's normality assumption and its dependence on three-seed effect estimates are stated there. The extension is reported separately from the registered three-seed runs.

Contextual LBF six-component confirmation

Base benchmark `Foraging-5x5-2p-2f-coop-v3` (lbforaging 2.0.0) with unchanged dynamics, local observations, six discrete actions and sparse collection reward. A wrapper balances two unlabeled geometric contexts (food identity 0 or 1 nearer to the team; two interior layouts per context selected by episode seed; horizon 15); the context label is never given to any controller. Training: the existing PPO implementation, 8,000 episodes per policy seed, no shaping. Trigger: food 0 collected first; basins: $\{\text{win}, \text{loss}\} \times \{\text{food-0-first}, \text{food-1-first}\}$; value: discounted mean sparse team reward ($\gamma = 0.98$); do(trigger)/do(non-trigger): minimal shared cooperative path/load planners toward the corresponding food, released once the target is consumed. Thresholds are copied unchanged from the previously frozen refined criterion. Evaluation: 80 episodes per context and condition; the training seed is the population unit. Two disclosed design pilots preceded the freeze and are excluded; confirmation seeds 1101–1110 were fixed in `CONTEXTUAL_LBF_PREREGISTRATION.md` (code SHA-256 recorded) before any confirmation training. The post-confirmation extension (seeds 1201–1205), the one-at-a-time threshold-sensitivity rescoring and the hindsight-optimal single-signal audit are post-hoc analyses stored in `contextual_lbf_extension_analysis.json`, `contextual_lbf_threshold_sensitivity.json` and `contextual_lbf_single_signal_audit.json`.

Strength-gradient battery

Same benchmark dynamics; rewards defined per provenance (shaped: team $+2.0\times$[trigger action]; outcome-only: team $+4.0\times$[rescue outcome]); five seeds each; pattern probability $p_t = P(\text{sacrifice-rescue basin})$ under near-greedy checkpoint policies (200 evaluation episodes, Laplace-smoothed); $C_t = -\log_2 p_t$. Open-space reference: 20,000 uniform-random episodes. Provenance rarity: mean C_t over the first quarter of training. Predictions ST-1–ST-4 were frozen in the script docstring; the ST-3 suddenness ordering failed on the 20-point/60k-episode grid because both provenances complete acquisition inside one checkpoint interval, and a disclosed fine-grid follow-up (40 checkpoints over 10k episodes) confirms later discovery and higher pre-discovery rarity for outcome-only while suddenness remains unordered (`strength_gradient_battery.py`, `strength_gradient_fine.py`).

Dual observer contracts

The second contract was declared before running: horizon 12, undiscounted success probability as value, and speed-refined basins (first-food identity $\times$ completion within 10 steps) that still resolve the trigger. Any admissible contract must resolve the declared macro-structure; contracts that erase the trigger identity are excluded by the measurement contract itself. All 75 stored systems (15 policy seeds $\times$ 5 systems, confirmation plus extension) were re-scored with frozen thresholds from saved policy networks and reconstructed initialization twins; registered predictions DO1–DO3 and the five conservative learned-side flips are recorded in `dual_observer_contracts.py` and `dual_observer_contracts.json`.

Learned basins, rollout models and the contract ensemble

For the machine-partitioned-basin audit, a k-means partition ($k = 4$, fixed initialization per seed) is fitted on the pooled natural episodes of each seed's five systems using behavioural summary features (first-consumed-food one-hot, foods collected, normalized episode length); the identifiability rule (mutual information between cluster and trigger ≥ 0.1 bits) is checked, and only potential and specificity depend on the partition. The cross-fitted audit uses low-level features only (padded per-step agent coordinates, per-agent action histograms, per-step reward sequences), fits on even-indexed natural episodes, freezes the partition, scores odd-indexed natural and all do-episodes, selects $k \in [2, 8]$ by silhouette on the training half, and repeats over k-means, Gaussian mixture, Ward agglomerative (nearest-centroid assignment) and PCA-10+k-means. For the rollout-model audit, learned policies and initialization twins are re-scored under softmax temperatures 0.2 and 2.0 with all thresholds frozen; the epistemic-versus-behavioural rollout distinction is declared in Terminology, and the IR-2 miss is retained. The contract ensemble pools the seven stored frozen-threshold contracts read-only and reports per-system acceptance rates (`learned_basin_clbf.py`, `crossfit_lowlevel_basins.py`, `independent_rollout_audit.py`, `contract_ensemble_analysis.py`).

Harmful emergence, matched provenance and the 2-D surface

The learned-harmful construction trains agent 0 on an exploiter reward (r_0 plus a bonus when its partner is lured onto the decoy goal); both value functions (u_{private}, u_{team}) were declared before training, and predictions HE-1–HE-4 passed 5/5 seeds (`learned_harmful_emergence.py`). The matched-provenance battery compares a hand script, a tabular behavioural clone of it, and the shaped and outcome-only learners of the strength battery, with a disclosed design pilot and predictions MP-1–MP-5 frozen after it (`matched_provenance.py`). The 2-D phase surface derives per-cell verdicts over goal reward $G \in \{3, 7, 10, 13\}$ and latent-context weight $w \in \{0.10, 0.25, 0.40, 0.70\}$ in closed form before training, declares one fragile cell (predicted margin 0.0625), and trains 5 fresh seeds per cell (`phase_2d_prediction.py`).

Continuous profile and calibration

The profile record and its fixed aggregation live in `emergence_profile.py`: potential normalized by $\log_2 |B|$; causal magnitude as the do-law JS divergence (bounded by 1 bit); signed value as tanh of the do-contrast over a per-domain scale declared before assembly ($\sigma_V = 5$ gridworld return units, 0.1 CLBF discounted-reward units); acquisition as the clipped structure gain over the initialization twin; abruptness as one minus the normalized entropy of positive collapse increments (bounded, thinning-stable). The certificate margin (soft-min over standardized component margins) measures evidence strength, not phenomenon strength, and is reported separately. Calibration: `profile_calibration.py` (orthogonal generator grid; dose–response; the original cross-term rank rule failure is retained) and `profile_existing_systems.py` (read-only taxonomy assembly). "Strong/weak emergence" is deliberately avoided as terminology. Ranking stability across the five stored contracts and the early-profile predictive-validity battery (four outcome-variable boundary cells, five fresh seeds each, snapshot at 25% of training) are scored in `contract_ranking_stability.py` and `predictive_validity.py`; the RS-2 and PV-1..3 misses are retained.

Ground-truth generator and axioms. The six-knob generator (`generator_calibration.py`) fixes trigger probabilities $p_1 = \frac{1}{2}(1+s)$, $p_0 = \frac{1}{2}(1-s)$, a do-trigger basin law of declared concentration b, a value contrast v, a twin carrying fraction $1-q$ of the structure, a logistic collapse path of steepness a, and a perturbation retaining fraction r; dimensions are measured with the full finite-sample pipeline (4,000 episodes per condition, three measurement seeds). Predictions GC-1–GC-5 were frozen in the script docstring; GC-2's exemption-list error and the $R \leftarrow b$ structural coupling are retained as a miss with a disclosed follow-up. The eight record axioms are verified in `verify_record_axioms.py` (20,000-sample ensembles for nullity, boundedness, monotonicity, data processing and context sensitivity; value/provenance separability pinned to GC-4/5; abstention pinned to the world-model follow-up). The admissible contract family and the identification-interval report are defined in `THEORY.md` and computed in `contract_ranking_stability.py`. The convergent-validity battery (`convergent_validity.py`, fresh 9800-series seeds, CV-1 pass, CV-2/3 retained misses) scores component-to-matching- endpoint prediction with a pooled model and no per-cell refitting.

World-model closure

Tabular MLE models of $(s, a) \mapsto (s', \text{events}, r, \text{done})$ are fitted on $K \in \{200, 1000, 5000, 20000\}$ on-policy trajectories (five data seeds per K, three policy seeds); components are recomputed with the model replacing the simulator in probe rollouts. The registered error proxy was mean pairwise TV between the five models' natural-basin distributions; WM-1/WM-2 passed, WM-3 failed and is retained. The disclosed follow-up adds an incomplete-rollout fraction cut of 0.10; it catches all 20 mismatches but abstains at K=20000, where forced interventions still hit unseen state–action pairs in 39–50% of rollouts (`world_model_closure.py`, `world_model_closure_followup.py`).

Overcooked-AI confirmation

The public benchmark is used unmodified (`overcooked_ai`, commit-pinned). Twelve self-play PPO seeds (77001–77012) are trained on the mixed layout pair (`cramped_room`, `asymmetric_advantages`), $5{\times}10^6$ environment steps, shared parameters, community shaped rewards annealed to zero over 60% of training. The trigger is agent 0 potting first; basins are (first potter $\times$ delivery); contract A uses sparse team value at horizon 400, contract B success-indicator value at horizon 300. Controls per seed: initialization twin, scripted role pair, behavioural clone of the scripted pair, and an independently initialized untrained pair; thresholds are copied unchanged from the frozen criterion. The full protocol and predictions OC-1–OC-5 were frozen with a public external timestamp before any confirmatory seed was launched; two design pilots are disclosed in the preregistration (`OVERCOOKED_PREREGISTRATION.md`, `overcooked_confirmation.py`, `overcooked_aggregate.py`).

Terminology and the system boundary

Observer contract: the declared measurement setting—a basin partition $B = g(\tau)$ that resolves the candidate macro-structure, a value function, a rollout policy, an intervention set, a time resolution, and a system boundary. *Trajectory-law family*: the set of trajectory distributions $\{P(\tau \mid c, \mathrm{do}(w))\}$ over declared contexts and interventions. *Collapse* names three related statistics that the text keeps separate: retrospective conditioning (Eq. 1, bookkeeping only), the within-run contraction of $P_t(B \mid s_t)$, and the interventional divergence between do-laws (both measured). *Process proxy*: the four-component checkpoint instrument; its burst component measures acquisition velocity, not emergence magnitude.

The rollout model requires the same discipline: it is an *epistemic* choice when it approximates the fixed target policy's own conditional futures (rollout counts, horizons, mild decoding changes that preserve the behaving distribution), and it becomes a *behavioural intervention*—defining a different system—when it alters the policy's action distribution itself, as strong temperature inflation does. The rollout-model audit is scored under this declaration.

The system boundary deserves an explicit statement because endogeneity depends on it. The contract declares the system to be the policy with its learned parameters at measurement time; endogeneity holds when the trigger is selected by that system's own action distribution rather than forced by the measurement apparatus, and it is deliberately paired with the *measured* acquisition component, which compares the same system against its own initialization and therefore does not rely on a design flag. The pairing has a known boundary: a network supervised to clone a script would pass acquisition, and the framework then classifies it as adaptive *with respect to its declared training signal*—provenance is relative to the declared boundary, not an absolute property. Reward, architecture and curriculum shape every learned system; the criterion does not adjudicate those upstream choices and no claim depends on it doing so.

Statistics and internal protocol freezing

Sign tests are exact binomial. All headline effect sizes carry 20,000- resample percentile bootstrap 95% CIs over their natural independent units. For chess the position is the independent unit. For deep MARL, episode-level intervals and sign tests quantify intervention effects conditional on the three trained policies; policy-seed medians are reported separately, and the episode-level p values are not used as population-level inference over training randomness. Confirmatory stages use author-maintained protocols frozen before their reported measurements, with outcomes recorded in place. These records are not third-party registered reports and do not provide an external timestamp; they are supplied for auditability. The exception is the Overcooked-AI confirmation, whose full protocol, thresholds and predictions were frozen with a public external timestamp before any confirmatory seed was launched (`github.com/FanYixiang2000/emergence-prereg`, tag `v1.0-overcooked-prereg`, commit `8415e45`, 2026-07-18); the confirmation script refuses to run without an explicit flag asserting that this timestamp exists. A single-page audit (`PREDICTION_LEDGER.md`) indexes ∼259 frozen verdicts/checks with 46 recorded misses, stratified into four tiers— roughly twenty primary confirmatory hypotheses (Tier 1), the replication/transfer hypotheses (Tier 2), diagnostic and mechanistic predictions (Tier 3), and software/artifact invariants (Tier 4); the main text's claims rest on Tiers 1–2 only, and the count is a transparency record, not a sample size—each row pointing to its primary document. Ledger rows are heterogeneous and dependent, so the count is not treated as a sample size or combined significance test. Reported p values test their local contrasts; no family-wise significance claim is made across the heterogeneous ledger. When a miss motivated a refinement, the miss remains counted as a miss; the revised component or threshold is recorded as a method change and contributes only through later frozen tests on fresh seeds, public checkpoints, or new task families. Code and outputs sufficient to regenerate every figure are provided (`REPRODUCIBILITY.md`).

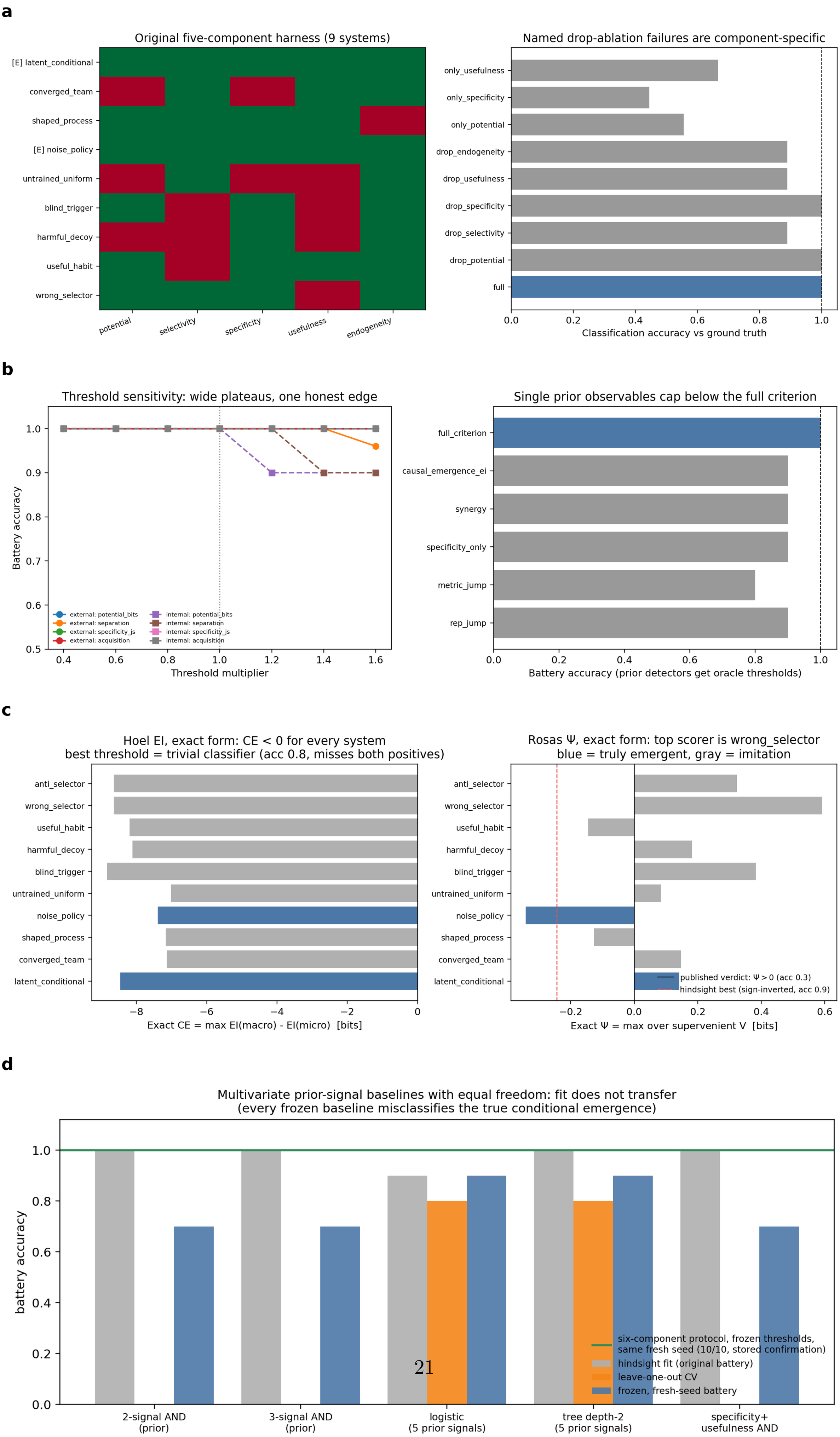
a
Original five-component harness (9 systems)
[E] latent_conditional
converged_team
shaped_process
[E] noise_policy
untrained_uniform
blind_trigger
harmful_decoy
useful_habit
wrong_selector
potential
selectivity
specificity
usefulness
endogeneity
Named drop-ablation failures are component-specific
only_usefulness
only_specificity
only_potential
drop_endogeneity
drop_usefulness
drop_specificity
drop_selectivity
drop_potential
full
Classification accuracy vs ground truth
b
Threshold sensitivity: wide plateaus, one honest edge
Battery accuracy
Threshold multiplier
external: potential_bits
external: separation
external: specificity_js
external: acquisition
internal: potential_bits
internal: separation
internal: specificity_js
internal: acquisition
Single prior observables cap below the full criterion
full_criterion
causal_emergence_ei
synergy
specificity_only
metric_jump
rep_jump
Battery accuracy (prior detectors get oracle thresholds)
c
Hoel EI, exact form: CE < 0 for every system
best threshold = trivial classifier (acc 0.8, misses both positives)
anti_selector
wrong_selector
useful_habit
harmful_decoy
blind_trigger
untrained_uniform
noise_policy
shaped_process
converged_team
latent_conditional
Exact CE = max EI(macro) - EI(micro) [bits]
Rosas Ψ, exact form: top scorer is wrong_selector
blue = truly emergent, gray = imitation
published verdict: Ψ > 0 (acc 0.3)
hindsight best (sign-inverted, acc 0.9)
Exact Ψ = max over supervenient V [bits]
d
Multivariate prior-signal baselines with equal freedom: fit does not transfer
(every frozen baseline misclassifies the true conditional emergence)
battery accuracy
six-component protocol, frozen thresholds, same fresh seed (10/10, stored confirmation)
hindsight fit (original battery)
leave-one-out CV
frozen, fresh-seed battery
2-signal AND (prior)
3-signal AND (prior)
logistic (5 prior signals)
tree depth-2 (5 prior signals)
specificity+ usefulness AND

Figure 3: The full six-component criterion confirms in three system families under frozen thresholds. **a**, External swarm: refined out-of-sample confirmation, 25/25 fresh verdicts across five policy seeds, with measured acquisition against initialization twins. **b**, Contextual Level-Based Foraging: 9/10 fresh learned policies pass all six components (retained miss: seed 1104 selectivity $0.4875 < 0.5$); 40/40 initialization and scripted controls reject; the competent team-nearest controller fails exactly endogeneity and acquisition. **c**, Latent-context sequence model (causal transformer, next-token training only): 10/10 fresh seeds accepted, 40/40 controls rejected (left); per-seed component values clear every frozen threshold (right). Thresholds are identical across the three domains and were frozen before each confirmatory run.

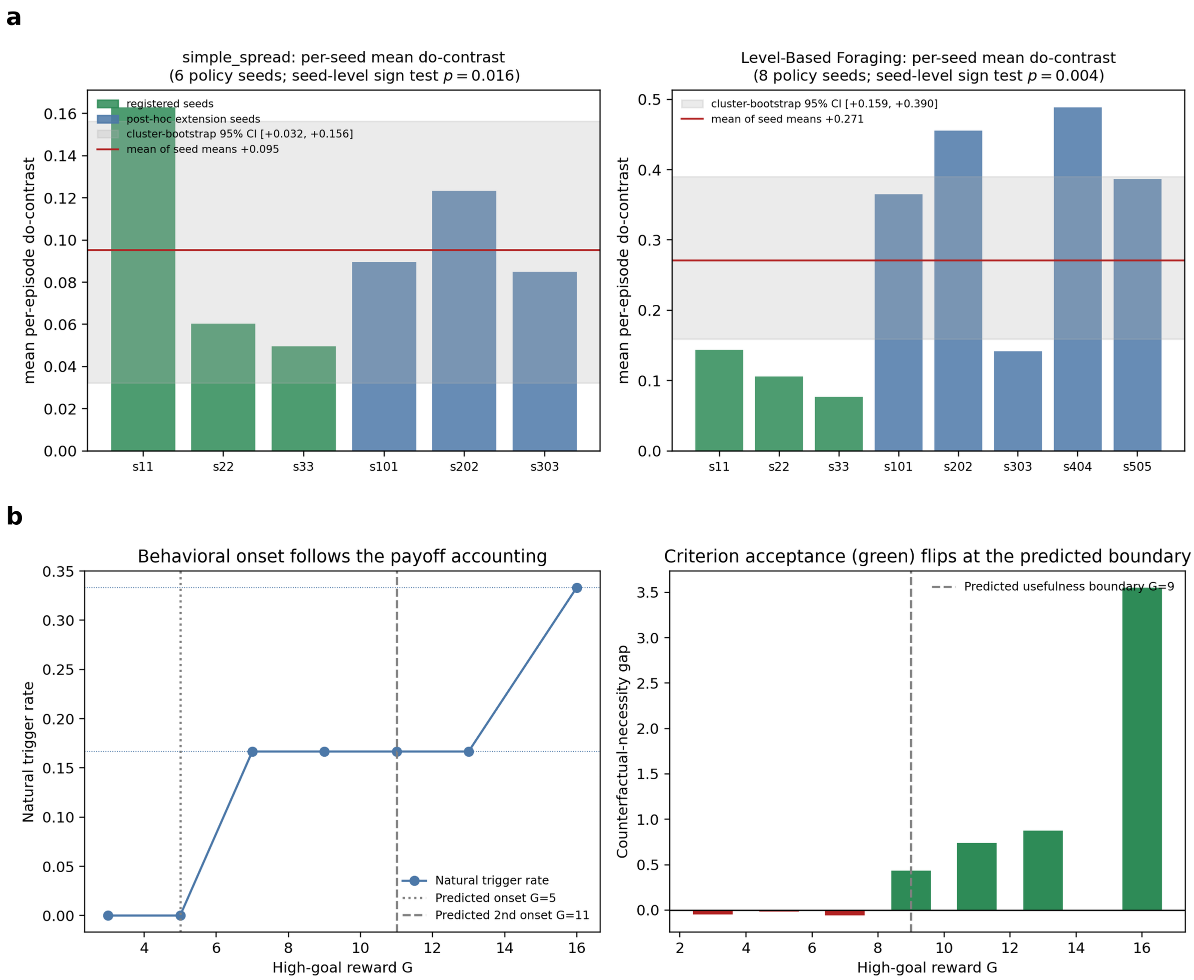

Figure 4: Deep multi-agent RL: seed-level inference and within-family parameter extrapolation. a, The training-population unit is the policy seed. Per-seed mean do-contrasts are positive for all six simple_spread seeds (seed-level sign test $p = 0.016$; cluster-bootstrap mean interval $[0.032, 0.156]$) and all eight LBF seeds ($p = 0.004$; $[0.159, 0.390]$); registered and post-hoc extension seeds are distinguished by colour and never pooled silently. Episode-level conditional tests and control panels are in Extended Data Fig. 11. **b**, The prospective phase-boundary prediction, shown for the primary registered run: behavioural onset follows the payoff accounting derived before training (left); criterion acceptance changes around the predicted usefulness boundary $G = 9$ (right). Three independent-seed replications matched 11/12 scored non-tie points, with stable outer phases.

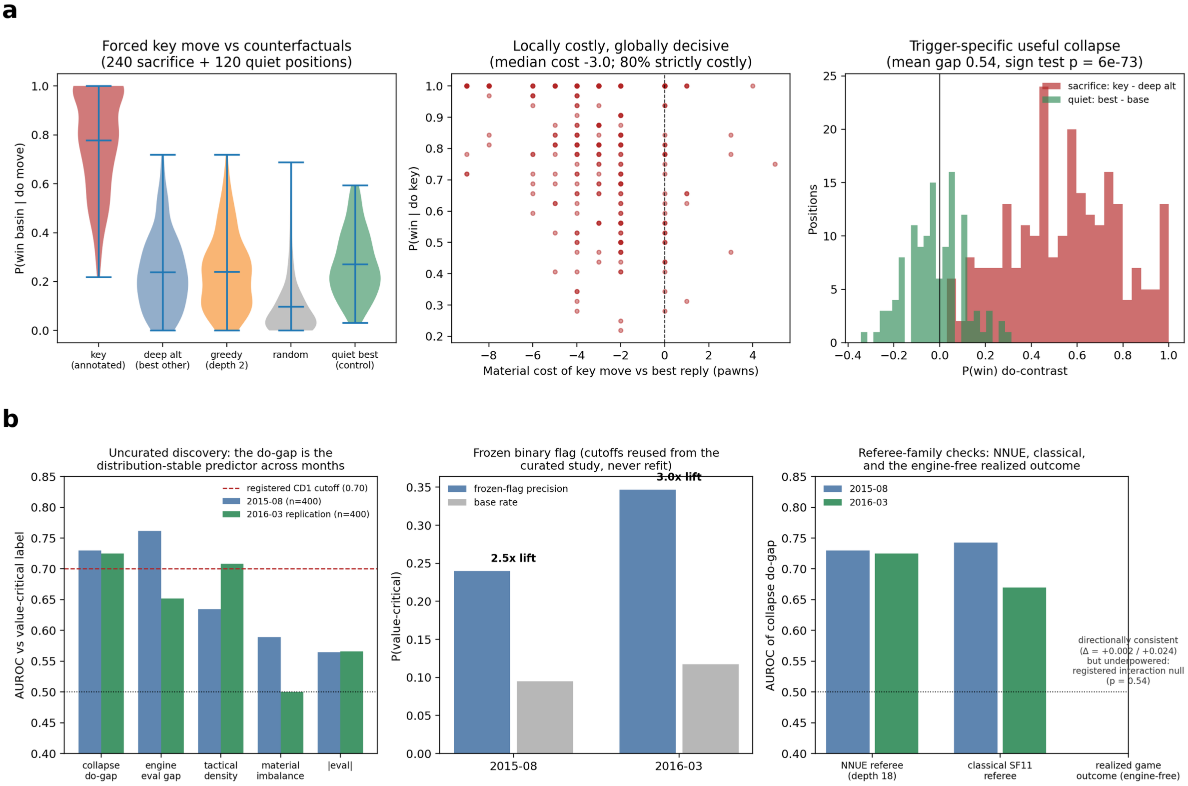

Figure 5: Chess: event-level recovery on curated positions and prospective discovery on uncurated months. **a**, Curated recovery: $P(\text{win-basin} \mid \text{do(move)})$ for the annotated key move vs. counterfactual alternatives across 240 sacrifice and 120 quiet positions (left); the key move is locally costly—median -3 pawns vs. best reply—yet globally decisive (middle); the do-contrast distribution separates sacrifices from quiet controls, mean gap $+0.54$ (right). **b**, Uncurated discovery under a separately frozen protocol: the collapse do-gap ranks value-critical decisions above tactical, material and eval-magnitude baselines in both months and is the distribution-stable predictor (left); the frozen binary flag lifts precision $2.5\times/3.0\times$ over base rate (middle); referee families—NNUE, classical SF11, and the engine-free realized game outcome—agree in direction, with the realized-outcome interaction registered as an underpowered null (right). Acquisition is not tested in this domain; estimator/engine robustness cells are in Extended Data Fig. 11.

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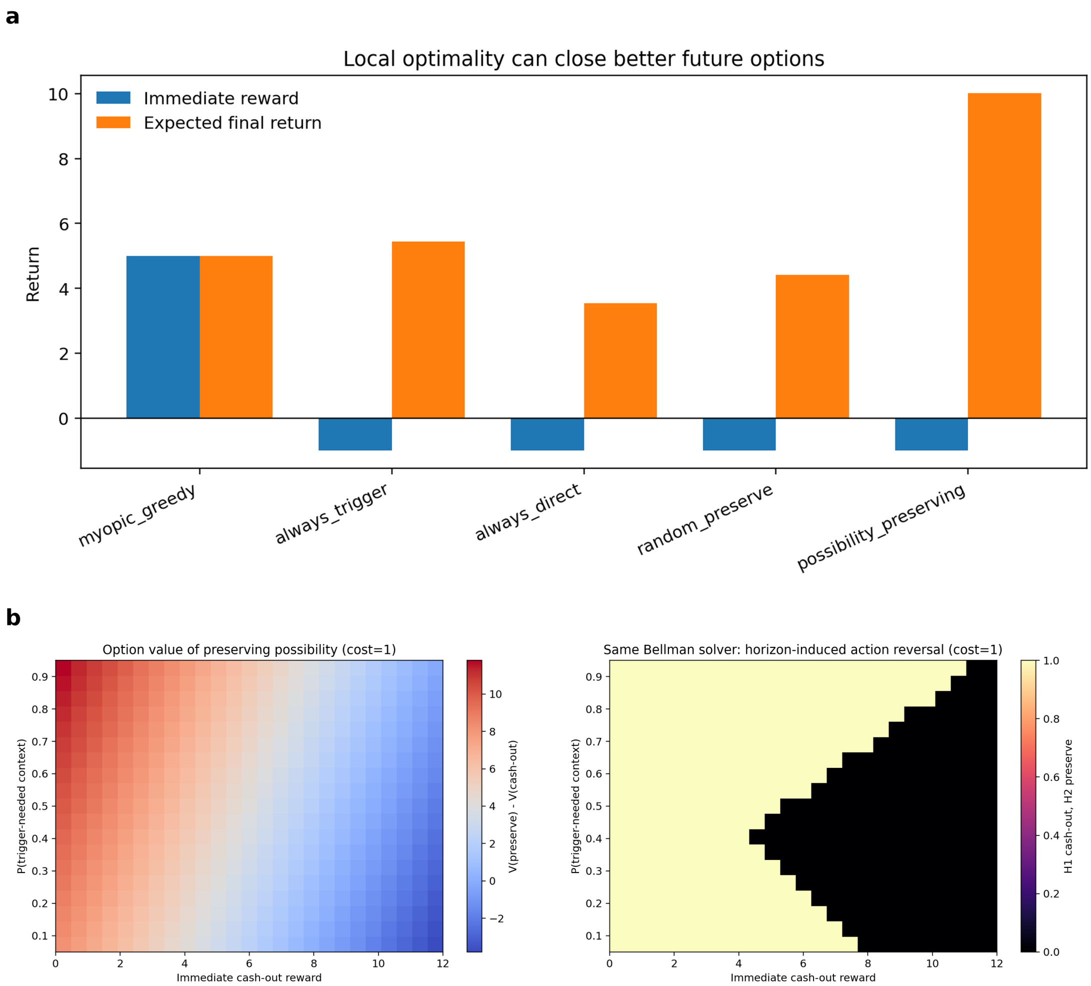
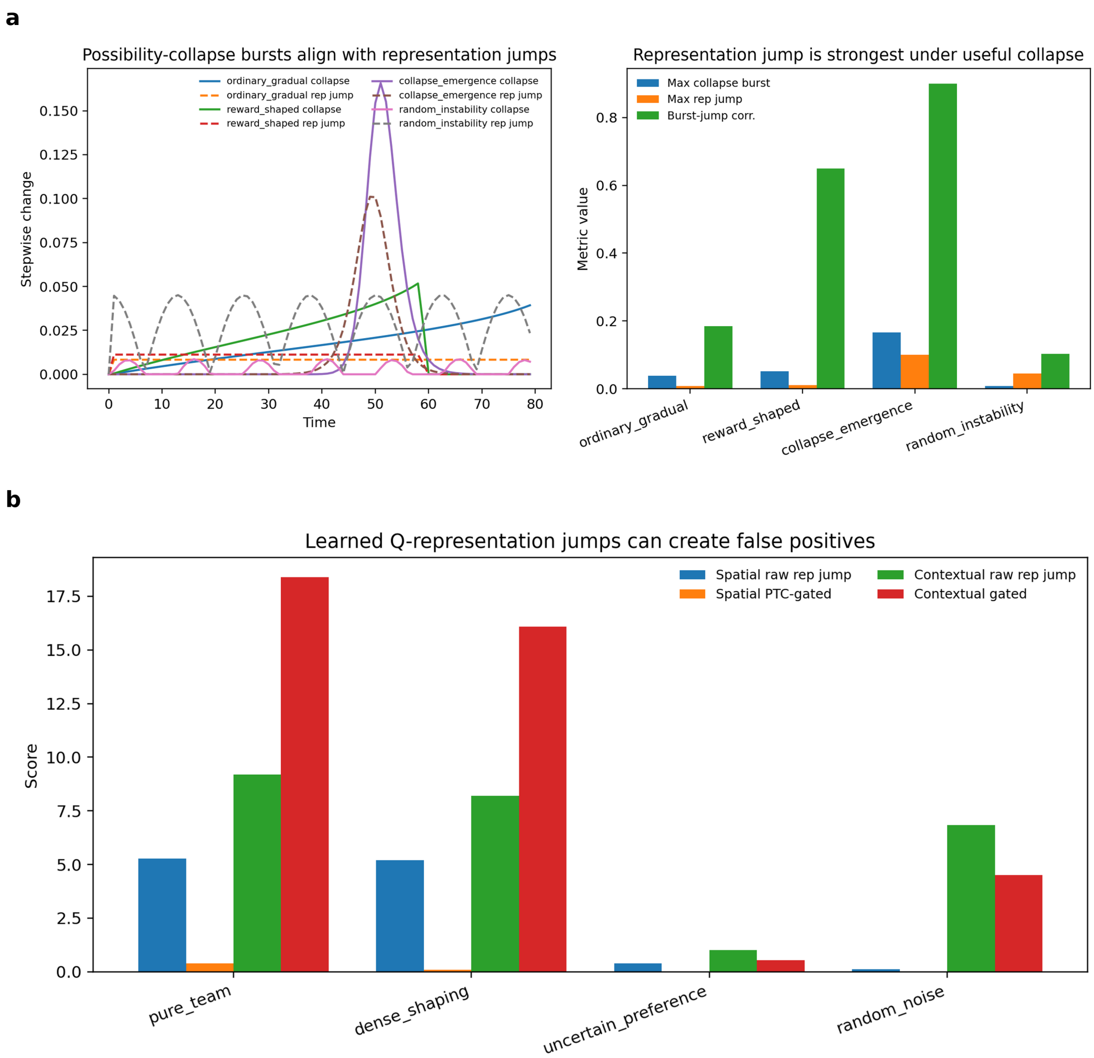
Extended Data

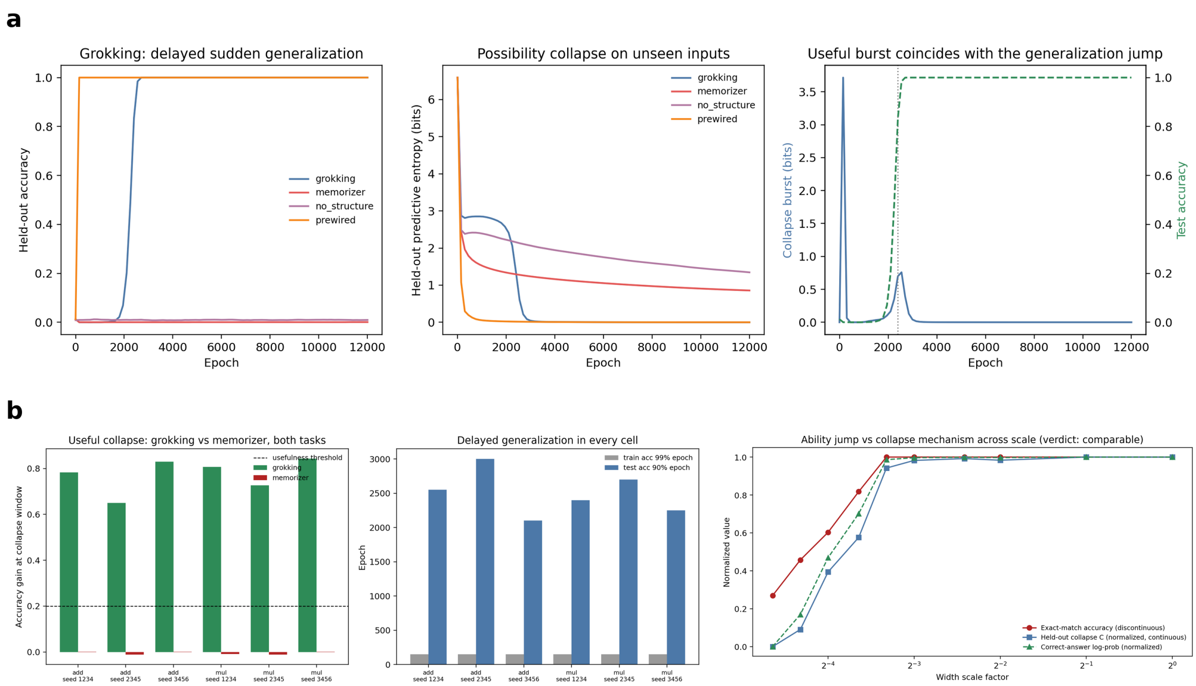
Extended Data Figs. 1–13 provide, in order: the analytic option-value core; the representation-jump bridge and false-positive audit; the grokking bridge, generality sweep and scale decomposition; induction heads and transformer grokking; the external swarm transfer; the estimator-robustness grid; process-grid robustness and hierarchical MARL inference; Pythia scaling; the within-episode mechanism; the public checkpoint series; episode-level MARL detail with the chess robustness grid; the dual observer contracts; and the emergence-strength gradient. The Contextual LBF full-criterion panels appear in main Fig. 3b. At submission, items beyond the Extended Data limit will move to Supplementary Information together with the full prediction ledger ($\sim$259 frozen verdicts/checks, 46 registered misses), the per-domain internally frozen protocols with outcomes recorded in place, the theory document with numerical verification of every proposition, and reproduction commands for every figure.



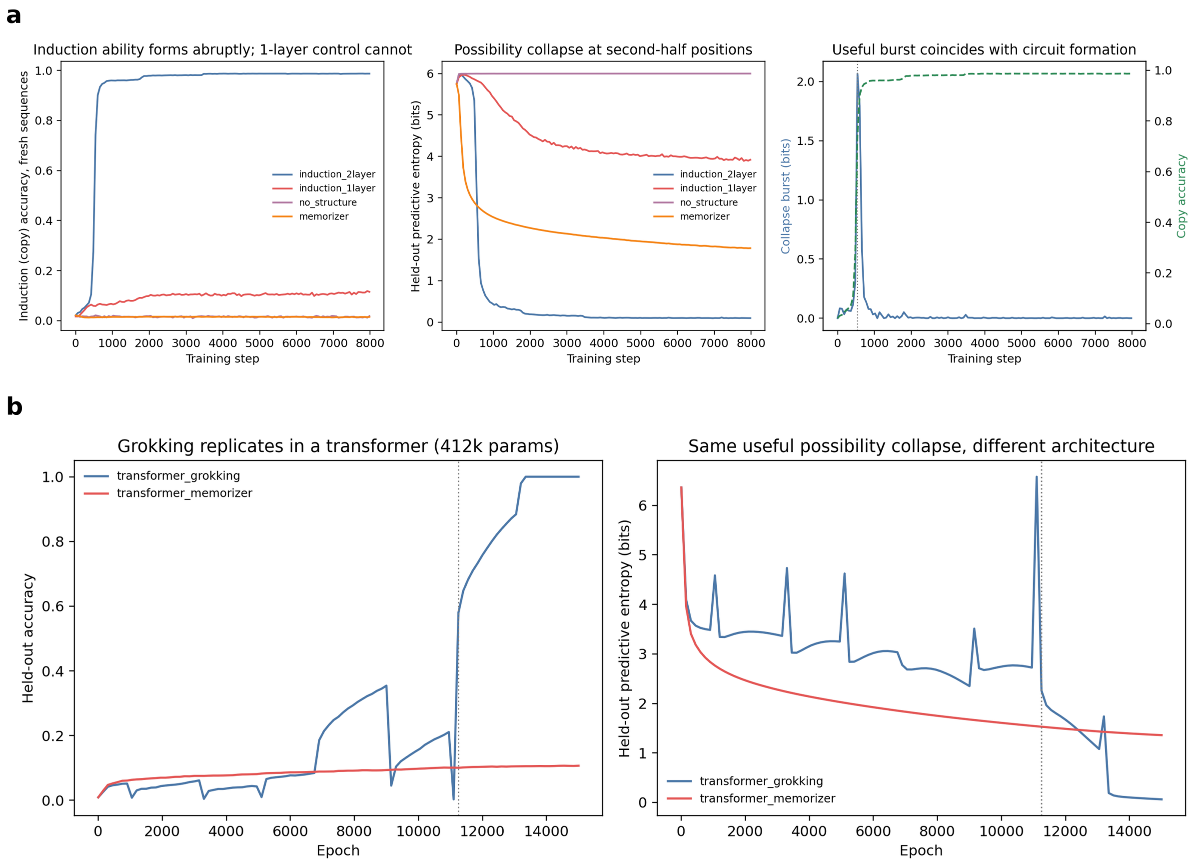
Extended Data Fig. 1: Analytic core: option value of preserving possibility. **a**, In the finite-horizon possibility tree, the myopically greedy policy maximizes immediate reward but closes the higher-return futures; the possibility-preserving policy pays a local cost and doubles the expected final return—local optimality can close better future options. **b**, The closed-form option-value boundary $V(\text{preserve}) - V(\text{cash-out})$ over context probability and immediate reward (left); the *same* Bellman solver reverses its action as the horizon expands (right): possibility preservation is a rational consequence of long horizons, not an added preference. These closed-form structures are what the phase-boundary prediction of Fig. 6c instantiates empirically.



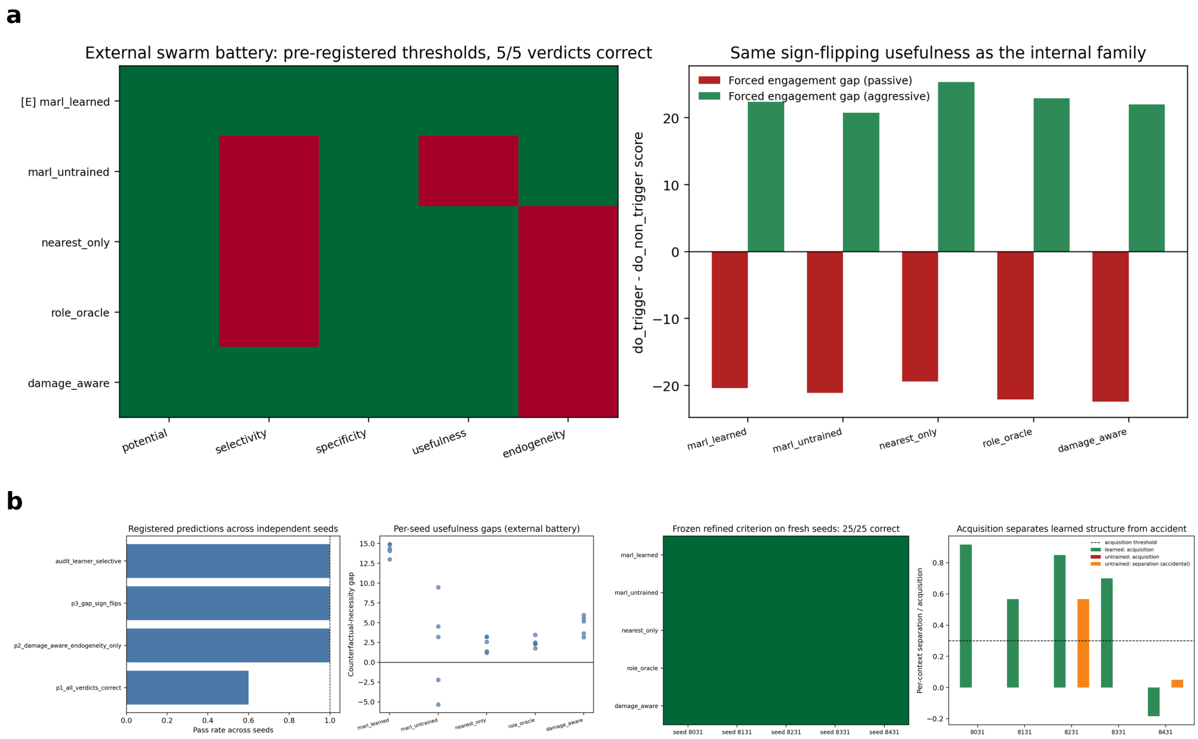
Extended Data Fig. 2: Representation jumps are a bounded symptom, not a definition. a, Training-time bridge across four regimes (collapse emergence, ordinary gradual, reward-shaped, random instability): collapse bursts and representation jumps co-occur in the emergence regime, and the Pinsker bound of Proposition 2 holds at every step of every trajectory. **b**, False-positive audit on learned representations: ordinary converged team reward and reward-shaped convergence produce jumps as large as (or larger than) true conditional emergence, and a noisy learner with the same acquired structure posts a 13.7× larger jump—unsigned distance measures noise, not emergence.



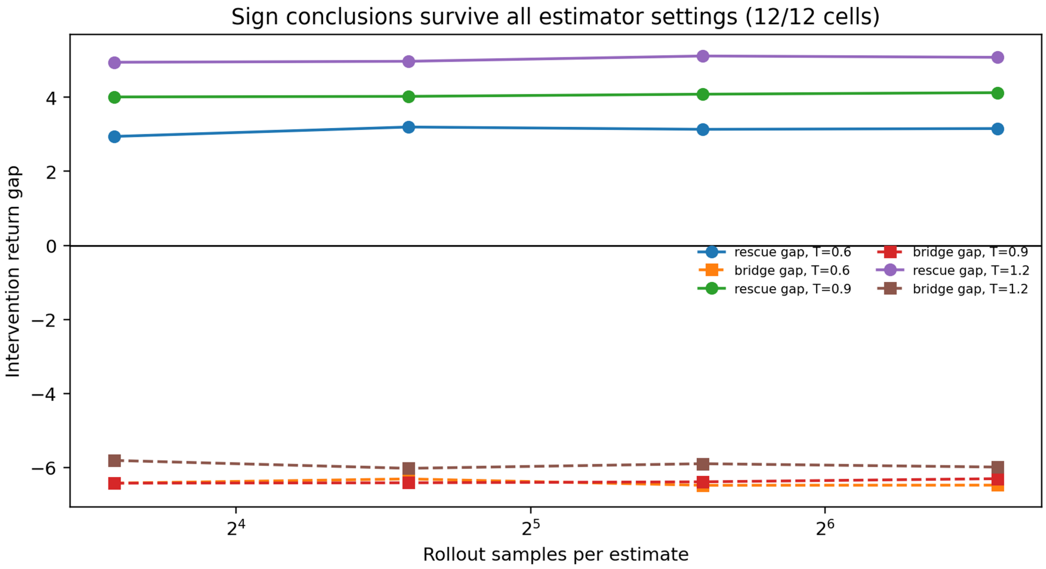
Extended Data Fig. 3: Grokking as useful possibility collapse. **a**, Modular-addition grokking (MLP family): the grokking run passes all process-level components; the memorizer fails potential, burstiness and usefulness; the no-structure control fails burstiness and usefulness; the prewired run (evaluation distribution included in training) is excluded *only* by endogeneity—the process-level analogue of reward shaping. The early memorization phase inside the grokking run itself is the sharpest internal control: held-out entropy drops $6.6 \to 2.8$ bits with zero accuracy gain, and the anchored window refuses to credit it. **b**, Generality sweep: 2 tasks $\times$ 3 seeds reproduce all verdicts (left); scale decomposition: the ability jump and the collapse transition land in the same scale region (right).



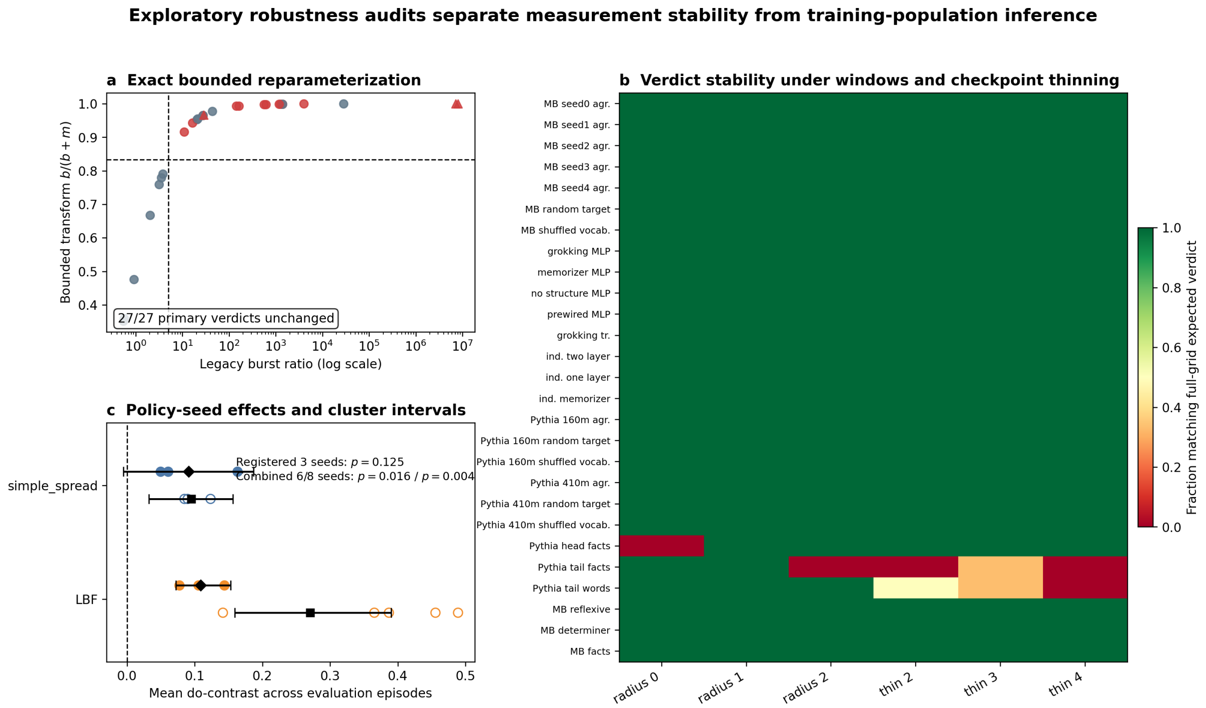
Extended Data Fig. 4: Externally discovered phase changes pass exactly when the architecture permits the circuit. a, Induction-head formation (attention-only transformers on cyclic-repeat sequences): the two-layer model—which can implement the induction circuit—passes all process-proxy components (copy accuracy 0.987); the one-layer model, provably unable to implement the circuit, fails usefulness (0.112); memorizer and no-structure controls fail as registered. 12/12 registered predictions on fresh seeds. **b**, Transformer replication of the grokking bridge: a 412k- parameter causal transformer grokks and passes; the same architecture without weight decay memorizes and fails usefulness. The process-level result spans MLP, causal-transformer, and attention-only families.



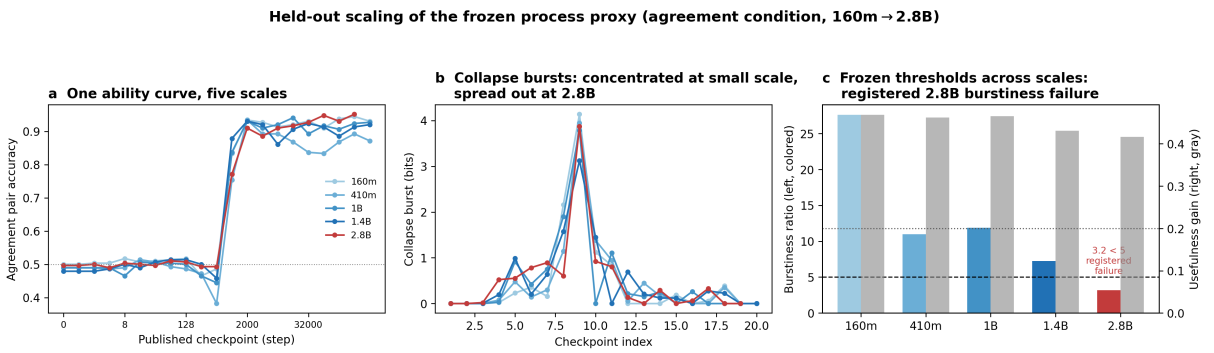
Extended Data Fig. 5: Prospectively frozen transfer to an external swarm benchmark. a, Five controllers on a continuous decoy/targeting swarm environment sharing no state space, action space, or dynamics with the internal family, thresholds copied unchanged: only the learned controller (audited engagement rate 1.00 in the aggressive context, 0.00 in the passive one) is accepted; the damage-aware controller—the external analogue of reward shaping—passes all four measured components and is excluded purely by endogeneity, pinning that component with an external counterexample. **b**, Multi-seed sweep with the registered failure (2/5 seeds accepted an untrained network on marginal tension; kept and reported) (left); the refined criterion (conditional selectivity, acquisition; frozen before new data) confirmed out-of-sample: 25/25 external verdicts on five fresh seeds (right).



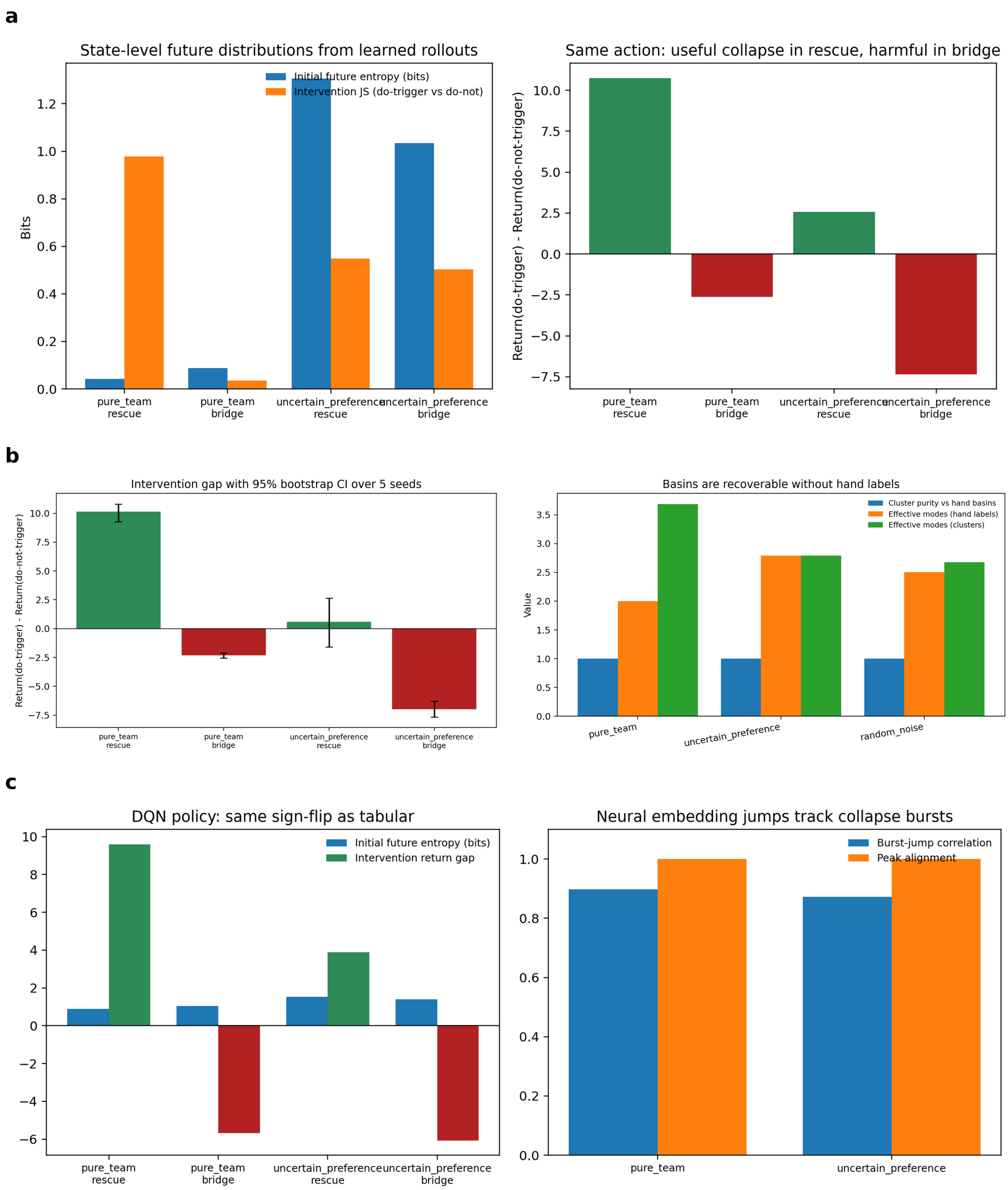
Extended Data Fig. 6: Estimator robustness. All qualitative conclusions of the within-episode probes (do-contrast signs, potential ordering) are unchanged across rollout counts $n \in \{12..96\}$ and probe temperatures (12-cell grid). Together with the 12-cell chess grid (Fig. 5b) and the LBF probe-temperature grid (Methods), this documents where the framework is estimator-robust and where its one declared observer-dependence lies: the *absolute* scale of the potential component.



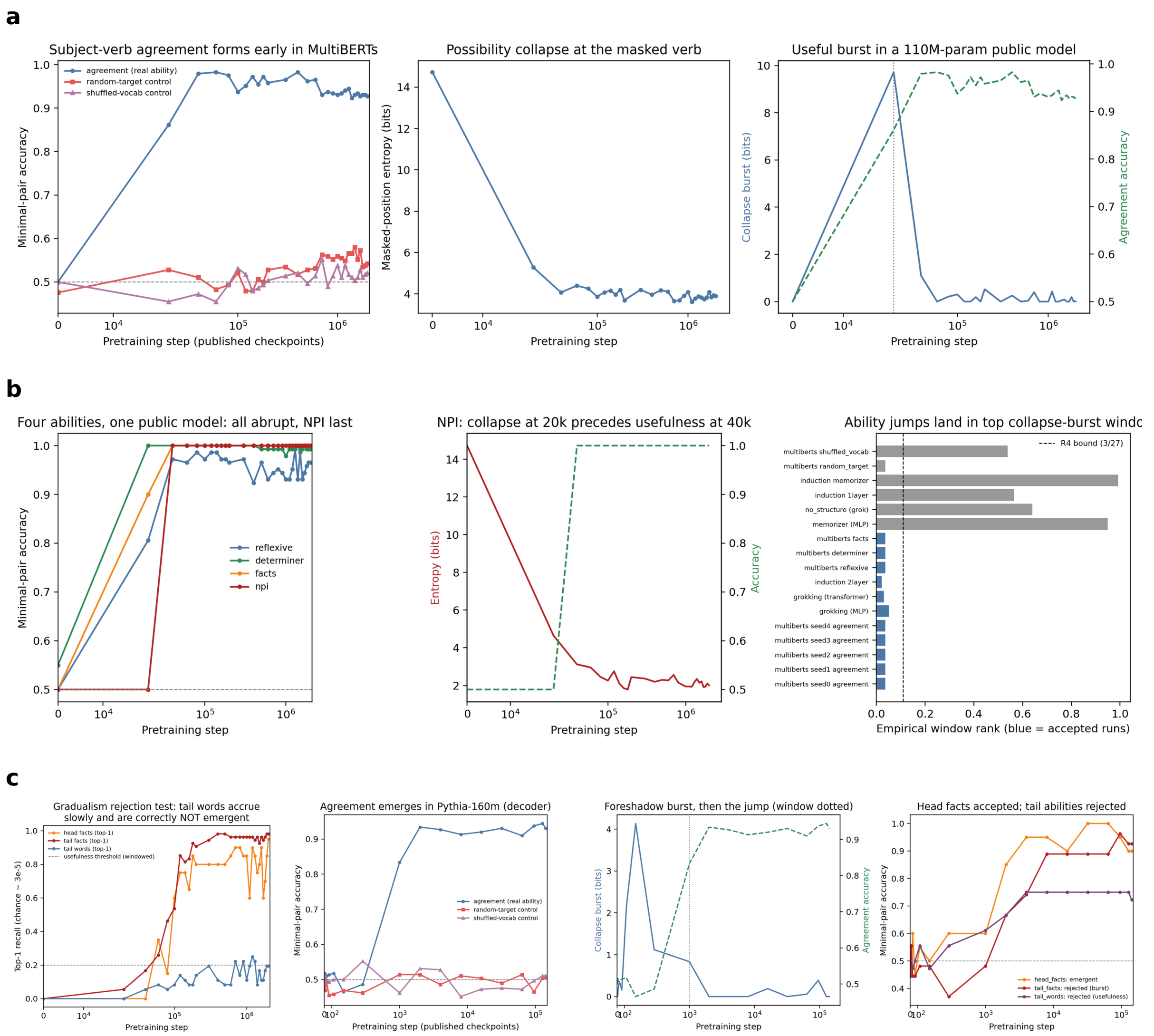
Extended Data Fig. 7: Exploratory process-grid robustness and seed-aware MARL inference. **a**, The bounded burst statistic is a monotone reparameterization of the frozen ratio; its threshold 5/6 is exactly the legacy ratio threshold 5, and 27/27 primary verdicts are unchanged. **b**, Verdict agreement under window radii 0–2 and checkpoint thinning factors 2–4. Full-grid agreement is perfect; 228/243 thinning cells agree, with disagreements concentrated in two Pythia tail controls. **c**, Per-policy-seed mean do-gaps for the three registered seeds (filled circles; diamonds/lines give their two-stage seed–episode bootstrap intervals) and for the post-hoc seed extension (open circles; squares/lines give the combined-seed intervals). With three seeds the exact positive-seed sign test is $p = 0.125$ in either domain and the simple_spread mean interval crosses zero; the extension (three more simple_spread and five more LBF policies, all positive) tightens these to $p = 0.016$ and $p = 0.004$ with positive combined mean intervals (Methods). All panels are post-hoc robustness analyses and are not counted as prospectively registered confirmations.



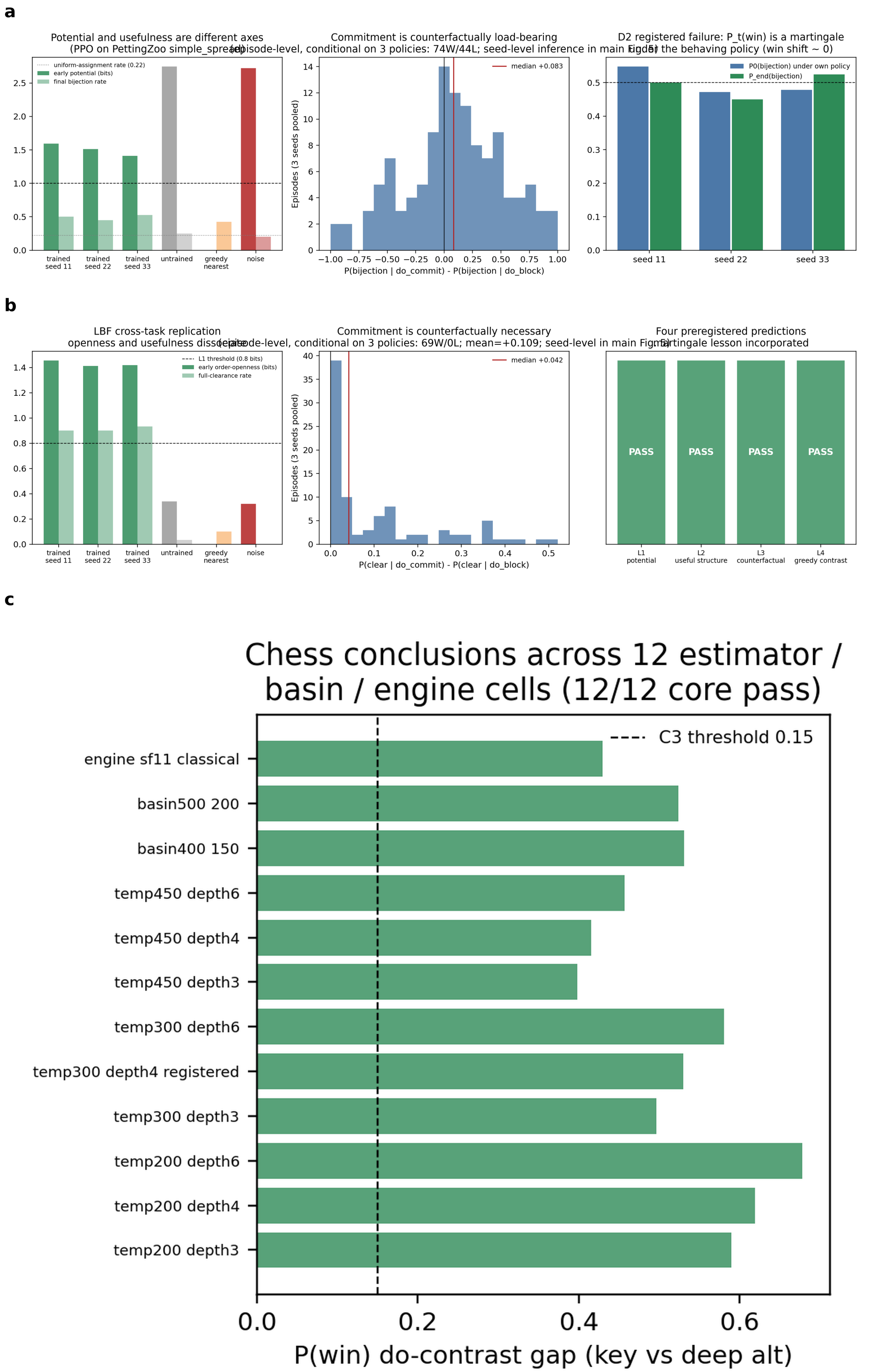
Extended Data Fig. 8: Held-out scaling of the frozen process proxy (Pythia agreement condition, 160m–2.8B). **a**, The same templated subject–verb-agreement battery under identical frozen thresholds at five public scales; every scale crosses from chance to > 0.85 within the same checkpoint region. **b**, Collapse bursts along the published checkpoint grid. At 160m–1.4B one dominant burst precedes the ability jump; at 2.8B the same total collapse is spread over several early intervals. **c**, Component values against the frozen cutoffs (dashed: burstiness 5; dotted: usefulness 0.2). Usefulness passes at every scale; burstiness declines monotonically with scale and fails registered prediction S1 at 2.8B ($3.2 < 5$), kept as a registered failure. Under every 2–4× checkpoint thinning the 2.8B verdict flips back to accept (registered S7 failure, $80.2\% < 90\%$ agreement), identifying the failure as grid-relative: larger models acquire the ability faster and more diffusely than the published log-spaced grid can resolve.



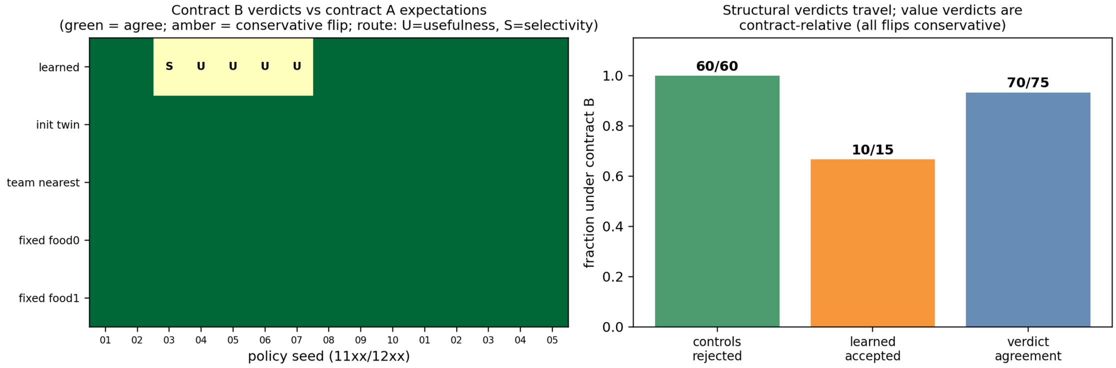
Extended Data Fig. 9: The mechanism within single episodes. **a**, State-level future distributions from learned rollouts: initial future entropy and intervention JS (left); the same action produces a useful collapse in the rescue context and a harmful one in the bridge context—the sign flip no unsigned signature carries (right). **b**, Intervention gaps with bootstrap 95% CIs over five seeds (left); basins are recoverable without hand labels: label-free clustering matches hand basins with purity 1.00 in the internal regimes and preserves effective-mode counts in the key uncertain-preference condition (right; pure-team loss episodes split more finely). **c**, Neural (DQN) replication shows the same sign structure (left), and neural embedding jumps track collapse bursts (correlation 0.87–0.90, peak alignment 1.0; right)—the representation jump as a downstream symptom, as Proposition 2 requires.



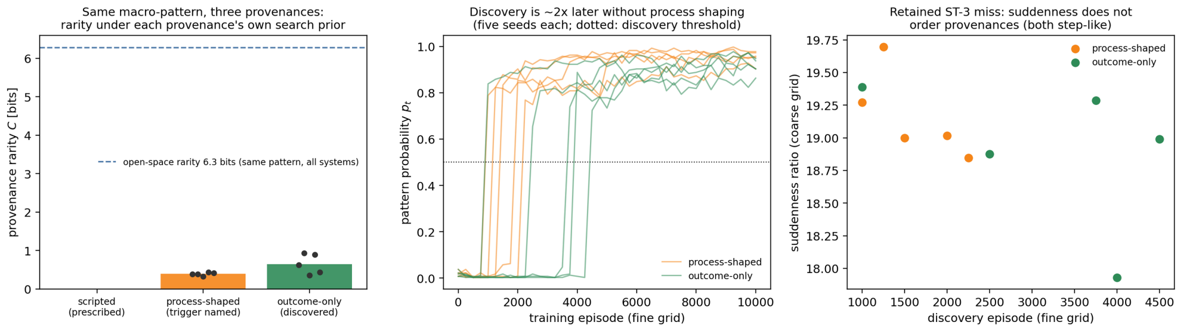
Extended Data Fig. 10: Training-time acquisition in public checkpoint series (no authorial control over training; observer protocol author-defined). a, MultiBERTs: subject–verb agreement accuracy jumps 0.50 → 0.98 while random-target and shuffled-vocabulary controls stay at chance (left); the masked-verb possibility space collapses from 14.7 bits (middle); the collapse burst coincides with the ability jump (right; agreement accepted 5/5 seeds and paired controls rejected, not 15 independent trials). **b**, Five-ability battery on the published checkpoint grid, NPI last (left); the NPI case shows collapse at 20k *preceding* usefulness at 40k (middle); a permutation test puts every accepted run's jump in a top collapse-burst window (right; descriptive per-run empirical ranks, not an omnibus significance test). **c**, The rejection side. Left: MultiBERTs tail-gradualism test—tail words accrue slowly and are correctly *not* emergent. Right three panels: Pythia decoder series—agreement emerges with both controls at chance; the foreshadow burst precedes the jump (dotted anchored window); head facts are accepted while *both* frequency-tail families are rejected, through two different components. Verified at two scales (160m, 410m).



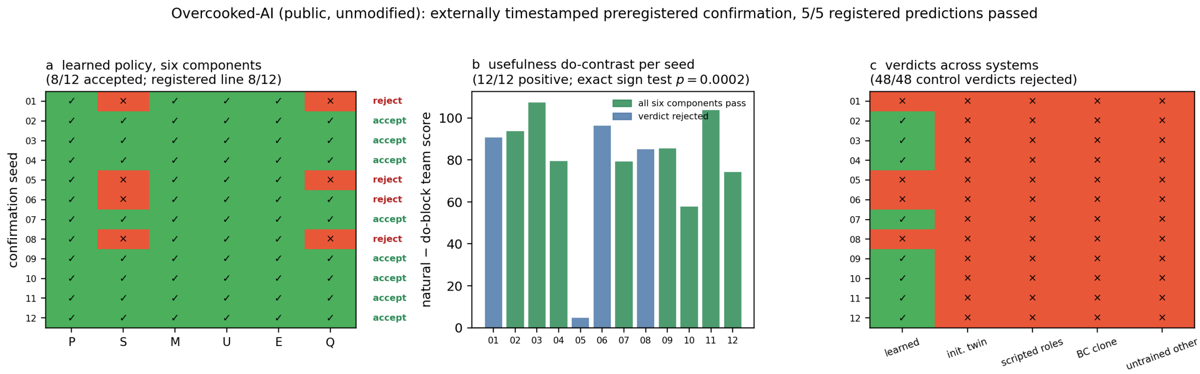
Extended Data Fig. 11: Episode-level MARL detail and the chess robustness grid. a, simple_spread (MAPPO): potential and usefulness are different axes—untrained openness does not qualify



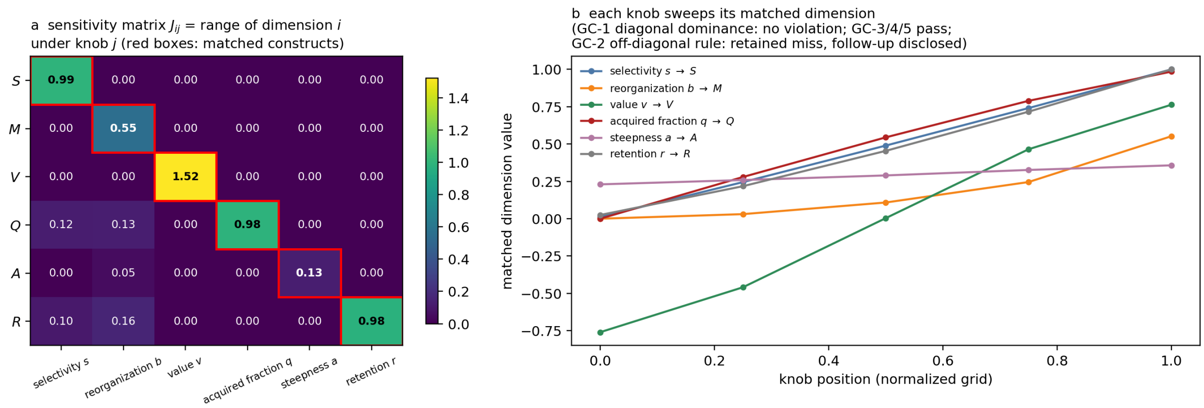
Extended Data Fig. 12: Two plausible observer contracts on the Contextual LBF systems. Contract B (declared before running) changes the horizon (15 to 12), the value function (discounted reward to undiscounted success) and the basin refinement (win/loss to completion speed) while still resolving the trigger identity. Left: verdict grid across all 75 stored systems; every disagreement (amber) is a conservative rejection of a learned seed, routed through usefulness (U) in four cases and borderline selectivity (S) in one. Right: controls are rejected under both contracts (60/60); structural components agree on 14/15 learned seeds; the value component is contract-relative, as the layered definition declares.



Extended Data Fig. 13: Emergence strength as provenance rarity: one macro-pattern, three provenances. Left: under the frozen open-space reference the sacrifice–rescue pattern is equally rare for every system (6.3 bits, dashed); under each provenance's own early search distribution the seed-mean rarity orders as the substrate predicts—scripted $0 <$ process-shaped $0.39 <$ outcome-only 0.65 bits (dots: individual seeds). Middle: fine-grid acquisition traces; the outcome-only provenance discovers the same pattern about twice as late as the process-shaped one, at matched final competence. Right: the retained ST-3 miss—collapse suddenness does not order the provenances, because both acquisitions are step-like at this grid; strength is carried by provenance rarity and discovery time, not burst shape.



Extended Data Fig. 14: Public third-party environment: externally timestamped, preregistered Overcooked-AI confirmation (all five registered predictions passed). **a**, Per-seed six-component outcome for the learned self-play policy across the 12 confirmation seeds: 8/12 accepted, exactly the registered acceptance line; every rejection routes through conditional selectivity (with acquisition), the correct verdict for policies that pot indiscriminately in both layouts. **b**, The primary seed-level inference: the usefulness do-contrast (natural minus do-block team score) is positive on 12/12 seeds (exact sign test $p = 2 \times 10^{-4}$); colour marks full-certificate acceptance. **c**, All 48 control verdicts are rejections: scripted role pairs and their behavioural clones fail endogeneity and acquisition despite high competence; initialization twins and untrained pairs fail selectivity and usefulness. The protocol, thresholds and predictions were frozen with a public external timestamp before any confirmatory seed was launched (Methods).



Extended Data Fig. 15: Six-knob ground-truth generator: the continuous record measures the constructs it names. **a**, Sensitivity matrix: response range of each measured dimension (rows) as each generator knob (columns) sweeps its grid with all other knobs at reference values, averaged over measurement seeds; every measurement uses the same finite-sample estimator pipeline as the real systems. Red boxes mark the matched construct–knob pairs; the registered diagonal-dominance prediction passed with no violation. **b**, Matched sweeps: each knob moves its own dimension across its full range (selectivity $0.01 \to 1.00$, value $-0.76 \to +0.76$, normalized acquisition $0.00 \to 0.99$, retention $0.03 \to 1.00$) while unmatched dimensions stay flat. Nullity (GC-3), value separability (GC-4) and provenance separability (GC-5) passed as registered; the off-diagonal suppression rule (GC-2) failed on two couplings and is retained—the frozen exemption list omitted the matched acquisition pair, and persistence responds to reorganization at 0.29 of its diagonal because retention is retention *of* structure.